CHAPTER XXXI

CONCLUDING REMARKS

Pitiless indeed are the processes of Time and Creative Thought and Logic; they respect the convenience of none nor the love of things held sacred; agony attends their course. Yet their work is the increasing glory of a world,—the production of psychic light,—the growth of knowledge,—the advancement of understanding,—the enlargement of human life,—the emancipation of Man. (264)

CASSIUS J. KEYSER

Yet the barbarians, who are not divided by rival traditions, fight all the more incessantly for food and space. Peoples cannot love one another unless they love the same ideas. (461)

G. SANTAYANA

The individual whose brutish desire for personal profit is unrestrained by the needs and rights of his fellows reverts to barbarism. If a bandit he is outlawed; if a politician he is—usually reëlected, with resulting retrogression of the entire social organization. (221)

C. JUDSON HERRICK

. . . a "League of Sound Logic" is the best "League of Nations" because effective under the subtle inevitable laws of Logical Fate—Unified Doctrines Will Unify Man. (280)

A. K.

A little less worry over the child and a bit more concern about the world we make for the child to live in; an inclusion of the child in a life of which the aim is not merely to earn money so as to become independent of the job; more love for whole-hearted, creative work and progress that will make possible what we all can share in; with these conditions, the adult and the young both will have a better chance.*

ADOLF MEYER

The present remarks were originally written for the last chapter of the whole volume, but a final critical survey of the material suggested the newly ordered sequence of the present three main divisions. Book I gives a general preparatory introduction which will help the reader to differentiate between the A and $\bar{A}$ systems, and to evaluate properly the differences. Book II formulates the main $\bar{A}$ principles which constitute an organic interrelated whole, to which the present concluding remarks belong. Book III gives some additional structural data about mathematics and physics which usually are not treated from the present point of view, but which furnish the essential structural material needed.

The writing of Book I, and particularly Book III, was very laborious and difficult. I often had the temptation to omit Book III entirely, and to refer the reader to other authors. After months of search, however, I found, to my sorrow, that, in spite of many excellent volumes, there were no books written from a structural and semantic

*What Can the Psychiatrist Contribute to Character Education? *Rel. Educ.* May, 1930.

point of view. To refer the reader to other writers would necessitate the reading of a fair-sized library, because often from a whole book he would need only a few scattered paragraphs. This would involve a very expensive and laborious process of hunting, which very few would undertake; besides, it would not give a connected structural or semantic picture. I tried to induce some specialists to write a book on the structure and semantic aspects of mathematics, and another similar book on physics. I was told that it would be very laborious and difficult, if at all possible, and so I had no other choice than to try to write it myself.

I earnestly suggest the reading of Book III, so that the reader may, at least, become acquainted with the existence of such problems. I hope that even specialists may find some suggestions helpful, because the structural and semantic aspects of science and mathematics are usually neglected, the neglect of which introduces needless difficulties in the teaching. The elimination of identification helps to solve many scientific puzzles, besides eliminating semantic blockages and so helping creative activities.

The world affairs have seemingly come to an impasse and probably, without the help of scientists, mathematicians, and psychiatrists included, we shall not be able to solve our urgent problems soon enough to prevent a complete collapse. Now those who are professionally engaged in human affairs, economists, sociologists, politicians, bankers, priests of every kind, teachers., 'mental' hygiene workers, and psychiatrists included, do not even suspect that material and methods of great general semantic value can be found in mathematics and the exact sciences. The drawing of their attention to this fact, no matter how clumsily done at first, will stimulate further researches, produce better formulations and understanding, and ultimately create conditions where sanity will be possible.

Some of those who have seen my manuscript or with whom I have discussed the problems seemed to dislike the term 'copying animals in our nervous reactions' and also the explicit introduction of 'Fido'. As identification is found among animals, primitives, infants, and 'mentally' ill, it could be said that the introduction of 'Fido' was not necessary. I have given serious consideration to the eventual desirability of completely eliminating 'Fido' from my work and substituting the term 'primitive'.; but, after mature deliberation, I decided that it will be helpful to accentuate the distinction between the reactions of animal and man. The main justifications of this are as follows:

1) My whole work and the formulation of a $\bar{A}$-system started with an attempt to produce a science of man, this necessitating a mod-

ern, scientific *functional, non-elementalistic, sharp* definition of man. Such a definition was given in my *Manhood of Humanity* to the effect that man differs from the animals in the capacity of each human generation to begin where the former generation left off. This capacity I called the time-binding function. This definition cannot be denied, and it fulfills the modern requirements.

2) The present enquiry originated in the investigation of the mechanism of time-binding, and is a further analysis of the sharp differences between the reactions of animals and humans, which became the psychophysiological foundation of a $\bar{A}$-system and a theory of sanity.

3) The further the investigation advanced, it became increasingly evident that the issues involved are extremely complex, and that in this field, from a structural and *non-el* point of view, practically nothing has been done. In general, all existing 'logics' and 'psychologies' are structurally misleading, since they are still thoroughly *el* and pre-A or A; these conditions necessitate the elimination of them, as well as other dependent disciplines, to prevent their being accepted as structurally fundamental. It was then desirable, in my pioneering enterprise, to keep a simpler and more obvious contrast between 'Fido', whom we nearly all know quite well and usually like, and 'Smith', whom no one seems to know properly. This method has proved very useful to the writer and I am convinced that many readers will find it equally helpful. I frankly admit that if I had not followed this simplified method, I could not have produced the $\bar{A}$-system and discovered in this psycho-logical maze the blockages introduced into our *s.r* by identification, elementalism, lack of consciousness of abstracting, improper evaluation., and, in general, infantilism.

For these three main reasons it seemed advisable to retain 'Fido' as a most useful factor in my analysis, with all due apologies to 'Fido'.

I also admit that I did not realize the difficulties of the task and the magnitude of the undertaking. The last revision alone of the manuscript required more than a year. I am all too well aware to what extent the presentation falls short of my expectations and how much better it could have been written by some one more gifted, but the following rather unexpected developments sustained my courage.

1) Curiously enough, the principles involved are often childishly simple, often 'generally known', to the point that on several occasions some older scientists felt 'offended' that such 'obvious' principles should be so emphasized. Yet my experience, without any exception, was that no matter how much these simple principles were approved of verbally, *in no case* were they *fully applied in practice.* Slowly I understood that

we cannot train mankind in identification by all available means, which must prevent adjustment, and then live by non-identification. Thus, when non-identity is pointed out, even a moron will 'agree', or wonder at the silliness of an author who fusses about it; yet, because all of us were *trained* in a linguistic and semantic system based on identity, that infantile identification will unconsciously play havoc with all our *s.r* the rest of our lives, unless this semantic blockage is counteracted. Naturally, the 'simpler' a principle appears, to which we may pay lip service, but which we *never fully apply,* the more I became convinced that the discovery of new methods for the application of this simple yet neglected principle must be considered most important. Any reader may verify by himself to what extent identification introduces difficulties in his own life. In fact, the main difficulties we have can always be traced to some identification somewhere.

2) The experimental data of Doctor Philip S. Graven with the 'mentally' ill and those cases of semantic disturbances which, in the orthodox way, were not supposed to be un-sane, showed that a change from the A standards of evaluations involving identification to the $\bar{A}$ standards without identification often either brought about a complete semantic reconstruction of an individual, or semantic, expedient, and lasting 'cures'. This fact again impressed me with the genuine workability and so human importance of a $\bar{A}$-system. If the old 'impossible to change', 'human nature' can be 'changed' by the new simple psychophysiological methods, this again suggests that this new system, no matter how imperfect, may be useful.

3) I was also very much impressed by the far-reaching power of the $\bar{A}$ methods. As a rule, only mathematicians and epistemologists fully appreciate what the power of a method means. Thus, the differential methods were invented, and later we found that these methods were structurally applicable to all processes. A tensor calculus was invented, and we found that it gave us absolute, invariant formulations applicable to all physics. Many other methodological innovations could be cited, and always the generality of the applications gave the value to those new formulations. The present $\bar{A}$-system was formulated in a way independent of other diciplines, as it was the direct result of structural semantic researches *free from identification.* This led to the formulation of fundamental *general* principles which underlie all human 'knowledge', such as non-identity, requiring the recognition of structure as the only possible content of 'knowledge' and so leading to the formulation of 'similarity of structure'; non-elementalism as a general principle; the general principle of uncertainty; ∞-valued general semantics,. It is

naturally very reassuring to find that the newest most important achievements of science have followed these principles unconsciously and have applied them *before* they were explicitly formulated.

From another point of view, a $\bar{A}$-system which could claim to be 'modern', should formulate general principles that all scientists in every field could follow. This was practically the case with the A-system until Francis Bacon. It is also the case with the present system except that different scientists have applied these new principles without having produced a *general formulation.* The fact that these principles had no general formulation was a retarding factor even in science and made the application of science to human affairs impossible. In the following examples, the different $\bar{A}$ aspects overlap, and I am emphasizing only the most marked features. Thus Einstein-Minkowski's space-time, Einstein-Mayer's new unified field theory, the newer quantum mechanics, the new physics of high pressure, piezochemistry., the tropism theory of the late Jacques Loeb, the physiological gradients of C. M. Child., ., exhibit clearly the application of non-elementalism. Heisenberg's restricted principle of uncertainty is also the result of the application of non-elementalism, based on the observation that the 'observer' and the 'observed' cannot be sharply divided. This principle becomes a particular instance of the general $\bar{A}$ principle of uncertainty, which again is based on the observation that we deal actually with absolute individuals and speak in more or less general terms, with the result that all statements are only probable in different degrees.

The absolute individuality of four-dimensional events, objects, situations, *s.r.*, necessitates an indefinitely flexible evaluation requiring ∞-valued $\bar{A}$ semantics. Outside of daily life, the best examples are given in science by the newer developments in vitamins, the effects of radiant energy on heredity, but particularly by the bewildering possibilities disclosed by the developments of physics, physics of high pressure, piezochemistry, polymorphism, colloidal behaviour, and the application of colloidal knowledge to psychiatry. The Polish school of mathematicians has produced the extension of the traditional two-valued A 'logic' to three-, and many-valued 'logic'; Chwistek has based a new foundation of mathematics and a new theory of aggregates on his semantic methods; but even these writers disregarded the *general* problems of non-elementalism, non-identity, and the necessity for a full-fledged $\bar{A}$-system before their formulations can become free from paradoxes, valid, and applied to life.

All these issues combined are of particular interest to mankind in general, and to the medical profession in particular, because, obviously,

if mankind is to pass from an infantile stage of its development into an era of general sanity, this would require a serious collaboration of medical science. Unfortunately, medical science is one of the most laborious and difficult disciplines, and, of late, in spite of some specific advances, it is rapidly ceasing to be in general a modern science. Any one who attends medical congresses, scientific meetings, or follows up medical literature often wonders whether he listens to, or reads, scientific arguments, or sixteenth century religious disputes. Dr. F. G. Crookshank, in his chapter on *The Importance of a Theory of Signs and a Critique of Language in the Study of Medicine* in Ogden and Richard's *The Meaning of Meaning,* gives an excellent picture of the present sad state of affairs; but a $\bar{A}$ analysis discovers deeper foundations underlying the difficulties in medicine, which would have to be remedied by the revision of medical education. In this connection, $\bar{A}$ issues become very important. Organisms in general, and humans in particular, represent colloidal processes which involve tremendous pressure because of colloidal attraction for water. Dr. Neda Marinesco[1] has recently suggested that Ice VI constitutes an important factor in the human organism. Ice VI represents a new form of ice discovered by P. W. Bridgman[2] who found that water in bulk and at the temperature of the body may be found to crystallize by the application of high pressure. It is the notion of Dr. Marinesco that the forces of adsorption may be as high as the pressure used by Bridgman, so that in thin surface films the arrangement of water molecules may be much like that found in Ice VI. It may interest the reader to know that, among others, Professor Bridgman discovered that aceton becomes solidified at room temperature, albumen coagulates., under high pressure.

Although physicians in their university days are well acquainted with colloidal chemistry, yet somehow, in practice, they have great difficulties in 'thinking' in colloidal terms. With the newest discoveries of physics of high pressure and piezochemistry, with their bewildering variety of physical manifestations, which, under different pressure, change with every individual material, a modern physician will have to 'think' not only in terms of colloids, but of colloids in combination with the data of high-pressure physics and piezochemistry. Now such 'thinking' is humanly impossible under the traditional two-, or three-valued A disciplines and becomes only possible with ∞-valued $\bar{A}$ general semantics. One of the immediate results of the use of $\bar{A}$ disciplines is the elimination of the elementalism of 'body' *and* 'mind', 'intellect' *and* 'emotions'., and the introduction of the *non-el* point of view as given in the present work. This requires every physician to be acquainted with

psychiatry, which acquaintance would eliminate many harmful cults. It should be fully realized that the older chemistry which dealt with different 'substances', having different 'properties', could have been treated by *A* subject-predicate and two-, or three-valued means. But not so in 1933; the older chemistry is gone, and today we deal only with a special branch of physics based on structure; the newer physics of high pressure show clearly that many of the older characteristics of 'substances' are only accidental functions of pressure, temperature, and what not, which vary in a bewildering way, requiring new semantic principles, new epistemologies.; in short, a new *non-el* and ∞-valued $\bar{A}$-system. In other words, whoever retains the *A s.r* is entirely unable to 'think' scientifically in the modern sense. If we want to have a science of man or a 1933 science of medicine., the first step is to revise thoroughly the *A-system.*

In fact, many more interconnections and interrelations could be shown which would make still more obvious how a $\bar{A}$-system results from, and leads to, modern scientific results, which can be extended and *applied to all human concerns* only after a *general* formulation *as a system.*

4) If the difference between the animal and man consists in the capacity of the latter to start where the former generation left off, obviously humans, to be humans, should exercise this capacity to the fullest extent. If we fail to do this, we again 'copy animals in our nervous reactions', which copying is the very thing we should struggle against. This 'where the former generation left off' would not only include all science, but also epistemology and the 'wisdom' which through painful experiences each former generation has accumulated., which, *in principle, should be given to every child.* Under the *A* conditions of our present education., systems, and evaluational systems, this is completely impossible and may sound visionary. Thus, to acquire scientific knowledge in all fields, one would have to spend a lifetime devoted to science, entirely free from financial worries., and even then he would only be able to acquire a small part of it. Before any older epistemological insight could be imparted, one should not only have special gifts, interests., but should also have an enormous amount of knowledge before such an education could be attempted. Similarly with 'wisdom'. The older and the younger generations, by colloidal necessity, cannot fully understand each other and, to a large extent, have mutual mistrust, which, as yet, is an entirely normal *A s.r*,.

In a *non-el*, $\bar{A}$-system this whole situation becomes radically changed. The impossible is made possible; I may say more, it is made

simple and easy, and becomes a necessary and unavoidable factor in the life of any child. A *non-el, Ā*-system is based on the complete elimination of identification, from which it directly follows that the only link between the un-speakable objective levels and the verbal levels is found in *structure*. Structure, then, becomes the only possible content of all knowledge., and all scientific technicalities, admittedly laborious and difficult, become only a necessary tool in the search for structure, with little, if any, intrinsic value, and are unnecessary for 'knowledge' as soon as in a given case the structure is discovered. This structure is always simple and can be given to children.

It is meaningless and utterly useless to argue whether or not the world is 'simple'; as the world *is not* our understanding of it; but as our 'understanding' happens to be structural, our nervous system, through its abstracting capacities, makes it simple, once its structural content is discovered. As the search for structure involves similarity of linguistic and empirical *structures,* we readily understand that any language, which we cannot evade teaching our children, has structure and involves structural assumptions. In the *structural revision* of our language and the teaching of a few structurally appropriate terms, entirely abandoning a few structurally misleading ones, we directly impart all up-to-a-date fundamental knowlege to any child. We train him automatically in the appropriate linguistic structure, which builds up in him appropriate *s.r.* Mankind at large does not need scientific technicalities to absorb and thereby obtain semantic benefits from the structural results of science. These results are the only ones which really matter, and which can be given in an extremely simple way, automatically abolishing the primitive metaphysics, structural assumptions, and infantile *s.r.*

By abolishing the structurally false to facts one-valued identification, we automatically train in ∞-valued differentiation leading toward consciousness of abstracting, which results in all the wisdom that epistemology and private experience can give us, being structurally a total result of racial experience. As structure is based on relations and *order,* structural training, when done consciously, becomes a physiological method, working simply and automatically.

In the *A*-system these semantic mechanisms were not consciously recognized, although they worked fatalistically with us. We were imparting primitive psychophysiological reactions to our children, who had to spend a lifetime to learn by very painful experience that something was wrong somewhere. Now we understand that the origin of the difficulty was in the lack of scientific investigations which would have analysed, non-elementalistically, the structural aspects of language and connected

s.r. All of which, let me repeat, works automatically, as experience and experiments abundantly show. Thus, an analysis of the mechanism of time-binding depends on the discovery of a sharp *non-el* difference between 'Fido' and 'Smith', and the formulation of means to make the time-binding characteristics of man fully effective with all except heavily pathological individuals.

By abolishing identification we generalize differentiation and so impart consciousness of abstracting, an indispensable factor in *proper evaluation,* and an absolute condition for adaptive and so survival behaviour. Thus a $\bar{A}$-system becomes a general theory of sanity and the general theory of time-binding, from which general semantics follow.

5) One of the most important features of the present $\bar{A}$-system consists of its *non-el* structural character. We may analyse problems in a scientific 'intellectual' way; yet this analysis, because *non-el,* structural, and semantic, appeals to, and affects, our 'feelings', 'intuitions'., involving psycho*physiological* factors based on order. Thus, the structurally necessary translations of one level of abstractions into the others and vice versa, is enormously facilitated, while in *el* systems these translations were hampered by unavoidable semantic blockages. Accordingly, 'intellect', 'emotions', 'body', and 'mind'., are not divided. The organism is affected *as-a-whole,* because structurally correct *non-el* means are employed, making many benefits of the system accessible to children, morons, and, perhaps, even superior idiots. The last results are to be foreseen, although they have not, as yet, been verified empirically.

6) But the most workable feature of the system consists in the fact that, being based on such fundamental principles as non-identification, non-elementalism., it has an organic unity. The main issues are all strictly interrelated and apply to 'body', 'mind', 'emotions'., in a *non-el* way, all working *automatically,* no matter from what angle we approach the training.

Thus, if we start with order, we are led to relations and structure; these establish differentiation and stratification, eliminating identification and 'allness', which result in consciousness of abstracting, necessitating ∞-valued general semantics, indispensable to proper evaluation and adjustment. If we start with non-identity, we are led to order, relations, structure, differentiation, stratification, non-allness, consciousness of abstracting, ∞-valued semantics, proper evaluation, and adjustment. If we start with differentiation or stratification, we are led to order, relations, structure, non-identification, non-allness, consciousness of abstracting, and proper evaluation.

It should be noticed that consciousness of abstracting and proper evaluation are complex end-results which cannot be imparted directly, but which become automatically lasting semantic states only after we have eliminated one-valued identification, or introduced order, ∞-valued differentiation, stratification , . The *non-el* benefit of the system consists in engaging the organism-as-a-whole. Thus, four-dimensional *order* plays the role of a potent *physiological* factor in the process and becomes the foundation for psychophysiology. Non-identity is a term applied on the verbal levels, which, on visual and intuitive levels, involves differentiation, ordering, and stratification. This system thus involves all necessary nerve centres and operates in a *non-el* way, as reactions on one level are easily and organically translated into the terms of other levels, making psychophysiology possible.

7) Finally, it is significant that many publications in the last ten years have shown efforts in a similar direction, which have received scientific and public approval. As I am more interested in creative work, rather than critical, I shall not analyse these strivings except to make one general remark that, because they are not based on order, structure, *non-el s.r*, the complete elimination of identification . , they are valuable and useful to the selected few, but under no conditions could a psychophysiology or a theory of sanity be based on these works which could be applicable in general elementary linguistic and semantic education. If I am not mistaken, in this respect the present work differs radically from the others with which I am acquainted.

From a *non-el* point of view we can never disregard the effect the 'body' or 'emotions' have on the 'mind', and vice versa the effect that the 'mind' has on the 'emotions' and the 'body', . Identification and all its consequences involve seriously disturbing semantic factors with corresponding colloidal disturbances, and it seems that, as yet, the human race, outside of very exceptional cases, has never been free from these disturbances. What effect the elimination of such disturbances will have on the human race it is impossible to foretell at this stage, beyond expressing the expectation that the consequences must be highly beneficial.

We have already become acquainted with the terms 'conditional' and 'unconditional' reactions. In the example of the patient and the paper roses, we have seen that the pathological symptoms were 'unconditional'. They were compulsory, as in the case of the dogs mentioned in Part VI. In a healthy individual they would have been fully conditional reactions, under semantic control. The above terminology may be extended so as to apply to all 'mental' ills, for here that which in the 'normal' person is a fully conditional reaction becomes unconditional, or a *reaction of*

lower order conditionality (compulsory) beyond conscious control. Here we differ from animals and hospital cases. When our conditional reactions are not fully regulated by proper *s.r* and become unconditional, we copy animals, and so are in a state of arrested development or of regression.

The general therapeutic and *preventive* measures are clearly indicated by such considerations. Conditional reactions in man should become *fully* conditional and not fixed as *un*conditional, or conditional of lower orders. In other words, instead of 'fixation', we should have means and methods to preserve and foster *semantic flexibility. This last is accomplished by acquiring the semantic reactions connected with the consciousness of abstracting.* I recommend this last point to the attention of specialists, as it is impossible in the present work to go further into details. Flexibility is an important semantic characteristic of healthy youth. Fixation is a semantic characteristic of old age. With the colloidal background, the imparting of permanent semantic flexibility which every one acquires who becomes conscious of abstracting might prove to be a crucial neuro-physico-chemical colloidal factor of, at present, unrealized power. The colloidal behaviour of our 'bodies' is dependent on electromagnetic . , manifestations, which, in their turn, are connected with 'mental' states of every description. If the colloidal ageing, which brings on old age, 'physical' and 'mental' symptoms, and, ultimately, death, is connected with such 'mental' fixity, we may expect some rather startling results if we impart a permanent semantic flexibility. The 'ageing' involves electrical changes in the colloidal background, which must be connected with the older semantic states. The new fluid semantic states should have different electrical influences, which, in their turn, would bring about a difference in the colloidal behaviour on which our 'physical' states depend.

From a $\bar{A}$ point of view, a new era of human development seems possible, in which, by mere structural analysis and a linguistic revision, we will discover disregarded semantic mechanisms operating in all of us, which can be easily influenced and controlled; and we will discover, also, that at least a great deal of prevention can be accomplished.

It seems, also, that we will discover more about the dependence of 'human nature' on the structure of our languages, doctrines, institutions . , and will conclude that for adjustment, stability . , we must adjust these man-made and man-invented semantic and other conditions in comformity with that newly discovered 'human nature'. This, of course, would require a thorough scientific 1933, physico-mathematical, epistemological, structural, and semantic revision of all existing human interests,

inclinations, institutions., to be made by those specialized in a 'science of man'. If such a revision is produced soon enough, it will, perhaps, help to adjust peacefully the standards of evaluation and prevent the repetition of bloody protests of unenlightened blind forces against *equally blind* forces of existing powers and reactions.

The forces of life, humanity, and time-binding are at odds; in modern slang, a 'show-down' is imminent; it *will happen,* and no one can prevent it. To a $\bar{A}$ understanding the only problem of importance is whether this 'show-down' will be scientific, enlightened, orderly, and peaceful, with minimum suffering; or whether it will take a blind, chaotic, silly, bloody, and wasteful turn with maximum suffering.

The problems of structure, language, and 'consciousness of abstracting' play a crucial semantic role. To be modern, one must accept modern metaphysics and a structurally revised modern language. As yet, these semantic problems have been *completely* disregarded as far as general education is concerned. This is probably due to the fact that in an infantile and commercial civilization we encourage engineering and applied sciences, medicine, biology., to increase private profits., and preserve or increase the ranks of buyers. But we do not encourage to an equal extent branches of science like mathematics, mathematical philosophy, linguistic, structural, and semantic researches., which would not directly increase profits or the numbers of customers, but which would, nevertheless, discover structural means for more happiness for all.

Accidentally—and this is recommended to the attention of economists—the classical law of 'supply and demand' is structurally and semantically an *animalistic law,* which in an adult human civilization must be reformulated. In fact, an adult human civilization cannot be produced at all if we preserve such fundamental animalistic 'laws'. In the animal world the numbers of individuals cannot increase beyond what the given conditions allow. The animals do not produce artificially.

Not so with our human world. We produce artificially because we are time-binders, and all of us stand on the shoulders of others and on the labours of the dead. We can over-populate this globe as we have done. Our numbers are not controlled by unaided nature, but can be increased considerably. In the animal world the numbers are regulated by the supply of food., and not by conditions imposed by the animals on that food supply. The animal law of 'supply and demand' is strict. In a human class of life, which does produce artificially, production should satisfy the wants of all, or their number should be controlled until the wants can be filled. The application of animalistic laws to ourselves makes conditions very complicated, and detrimental to most, if not all

of us. It is also easily understood why it should be so. Ignorant and *A* handling of powerful symbols has proved to be dangerous when we do not realize the overwhelming semantic role and the importance of symbols in a symbolic class of life.

Another interesting application of the consciousness of abstracting is given in our attitude toward money, bonds, titles to property,. Money represents a symbol for all human time-binding characteristics. Animals do not have it. No doubt bees produce honey, but these products of the bees do not constitute wealth until man puts his hands on them. Money is not edible or habitable. It is worthless if the other fellow refuses to take it. The *m.o reality* behind the symbol is found in *human agreement*. The *value* behind the symbol is *doctrinal*. Fido does not discriminate between the different orders of abstractions. If we copy him, we worship the symbol alone. 'In gold we trust' becomes the motto, with all its identifications and destructive consequences. Smith should not identify the *m.o* reality behind the symbol with the symbol. It is amusing, when not tragic, to see how the so-called 'practical man' deals mostly with fictitious values, for which he is willing to live and die. When he has the upper hand and ignorantly plays with symbols, disregarding the *m.o* realities behind them, of course, he drives civilization to disasters. History is full of examples of this.

We see the utter folly of racing to accumulate symbols, worthless in themselves, while destroying the 'mental' and 'moral' values which are behind the symbols. For it is useless to 'own' a semantically unbalanced world. Such ownership is a fiction, no matter how stable it may look on paper. Commercialism, as a creed, is a folly of this type. Some day even economists, bankers, and merchants will understand that such 'impractical' works as this present one on structure, *s.r.*, lead to the revision of standards of evaluation and are directly helping the stabilization of an economic system. Meanwhile, in their ignorance, they do their best to keep the economic system unscientific, and, therefore, unbalanced. History shows clearly how the rulers have generally made life unbearable for the rest of mankind, and what bloody results have followed. Since the World War certain conditions are becoming increasingly more difficult, and the infantile and animalistic systems drive us fatalistically toward further catastrophes. Whether these disasters will occur, the unknown future shall decide; but out of this unknown, one fact remains a certainty; namely, that this will depend on whether or not science can take hold of human affairs; I hope it can, but the blind forces of identification are so strong and powerful that perhaps such

hopes are premature. Perhaps a new race can accomplish it after this one is extinct, with the exception of a few remnants in museums.

The problems of determinism and indeterminism are not purely 'academic' but influence, to a large extent, our theories and behaviour, and so are fundamental for adjustment. Historically, science has utilized determinism of the two-, and three-valued variety, which has lately, in the case of the newer quantum mechanics, proved insufficient. The lack of the formulation of ∞-valued semantics, necessary for ∞-valued determinism, seemed to indicate that even science tends to drift toward indeterminism, a tendency which was rather baffling and disturbing to many scientists.

Different 'ethics' and 'morals' have fought determinism throughout all our past on the ground that in a deterministic world all 'morals' and 'ethics' would be impossible. If a man is compelled to do something, then, we are told, he is not responsible. They state that the result would be undesirable licence, forgetting that determinism implies quite the opposite of licence.

We have already become acquainted with infantile self-love and self-importance. These infantile characteristics have not only shaped our semantic attitudes, but also our 'scientific' theories. Smith and this little earth were in many ways postulated as the centre of the universe. Scientific discoveries showed that such statements did not cover the facts at hand, and Smith was displaced from this primitive and infantile self-centred position. The Polish astronomer Copernicus was the first to give this rude shock. The little earth was no longer *the* 'centre of *the* universe'. Next came Darwin with another shock to such infantile pride. Smith was no longer a 'special creation', but belonged to the general series of living forms, none of which were 'special creations'. Finally, Freud developed the notion that even in semantic processes, determinism prevails. All our actions, psycho-logical and semantic states . , have very definite conscious and unconscious psychophysiological 'causes' which activate us.

An infantile society had difficulties in abandoning their pleasing delusions, and these three men were duly persecuted, criticized, and bitterly attacked and hated by many.

The present situation may appear baffling because science discovers facts which would seem to lead to an 'undesirable' indeterminism in science, and to a determinism in 'mental' processes. The reader, by now, I hope, realizes that both 'undesirable' results are only undesirable because of identification and the confusion of orders of abstractions, which resulted in the ascribing of *undue generality and uniqueness* to the *A*

two and three-valued 'logic'. But once we realize that in a $\bar{A}$, ∞-valued, more general system, the two-, and three-valued aspects are only particular instances, which apply to some instances but not to others, all our difficulties vanish. From a $\bar{A}$ structural point of view we also understand that ∞-valued determinism becomes a necessity of our *s.r* in the search and comparison of structures.

The result seems to be that the problem of determinism or indeterminism is not primarily a problem of the outside world, but simply one of our *s.r* and ignorance versus 'knowledge'. Abandoning elementalism and identification, we stop arguing '*is* the *world* deterministic or not'.; but, by analysis, we find which semantics better fit, structurally, the facts and our abstracting capacities. The results we reach are not entirely new, but the semantic conflict is eliminated.

Science employs determinism because of the structure and function of our nervous system. We cannot do otherwise than preserve ∞-valued determinism and step by step supply the missing links in our structural adjustments of language to the structure of empirical data.

Let us again repeat that the older problems of 'determinism' *in general* were the results of elementalism and identification and of a complete misunderstanding of the role of structure. Once these undesirable afflictions are eliminated, the artificial problems which they create are also eliminated. Structural considerations show clearly that determinism is a neurological necessity. If empirical facts lead to linguistic indeterminism, it is an unmistakable sign that the language used is not similar in structure to the structure of the world around us, and that we should simply produce a language of different structure. *Such determinism is a vital condition in the search for structure, and cannot be abandoned.*

Shall we, then, preserve the deterministic attitude in our 'mental' processes? Are the objections on 'moral' and 'ethical' grounds serious enough to induce us to reinstate in our semantic attitudes the older structurally misleading 'indeterminism'?

Let us, first, recall the facts. In our old *el* and infantile attitudes with identification we analysed a child or an adult in *isolation*. Determinism was applied to such a fictitious *non-existent* individual, and the old objectified and *el* speculations followed. If any one is inclined to challenge the above statement, let him perform an experiment and immediately after birth isolate a child 'completely'. He will find that this cannot be done with a human baby without destroying the child. Therefore, the old speculations deal with structurally *fictitious* conditions. The facts are that a baby is, from the first, subjected to a treatment based

on the semantics, structure of language, doctrines, understanding, knowledge, attitudes, metaphysics., of his parents or their substitutes, which *shape his semantic reactions.*

If we abandon the problem of the two-valued 'determinism' in connection with such a fictitious, isolated individual, and apply ∞-valued determinism to an actual, non-isolated individual, we see at once that the whole situation is different. If parents and society accept ∞-valued determinism, they realize their own *responsibilities* toward the individual, and understand that the actions of parents, society., are, to a large extent, responsible for the future development of the child on quite deterministic psychophysiological grounds. If an individual behaves in a way detrimental to others and to himself, and an enlightened society decides to do this or that with him, that is a different proposition. The main point is that, if we were to accept an indeterministic attitude, a great deal of harm would be done by parents, teachers, preachers, and society in general; harm which could be prevented. This is, to a large extent, unrealized, and in the old way no one was supposed to be *responsible* except the poor victim of 'free will'. Under such *A* conditions, we sponsor bitterness, cruelty., under the labels of 'sin', 'justice', 'revenge', 'punishment', or whatever it may be. On deterministic grounds, when society and educators realize fully their own responsibilities, we should blame the individual less, and should more and more investigate structure, language, our systems, metaphysics, education, conditions of living,. Instead of a holy frenzy for 'justice', 'punishment', 'revenge'., we would try to improve conditions of life and education, so that a newborn individual would not be handicapped from the day of his birth.

Since the organism operates as-a-whole and *no one* is free from higher order abstractions and structural assumptions, we see that the *keeping of savage-made metaphysics* must involve us individually and collectively in an arrested or regressive development. From the organism-as-a-whole point of view structural ignorance must result in some semantic defectiveness.

The objection that there are cases of great 'mental' brilliancy accompanied by very vicious tendencies is easily answered by the fact that the problem is formulated in an *el* way. 'Mental' brilliancy does not tell the whole story of the organism-as-a-whole. One may be 'mentally' brilliant, yet infantile or a 'moral imbecile'. In life, we deal with the whole non-isolated individual, who may be pathological in a great many ways. If it is objected that science is so complicated that it would be impossible to impart such knowledge to the masses, the answer is that, as this enquiry shows, science involves some structural metaphysics and seman-

tic components which, once discovered, are childishly simple, and can be given in elementary education.

Science represents the highest structural abstractions that have been produced at each date. It is a supreme abstraction from all the experiences of countless individuals and generations. Since the lower centres produce the raw material from which the higher abstractions are made, and these higher abstractions again influence the working of the lower centres, obviously *some means can be devised to put back into the nervous circuits the beneficial effects of those highest abstractions.*

The above statement may appear visionary, and many are likely to say, 'It cannot be done'. Now, the main contention of the present theory, verified empirically, is that it *can* be done in an extremely simple way, provided we study the neglected *non-el* aspects of mathematics and science; namely, their structural and semantic aspects. Such study has helped us to discover in a $\bar{A}$-system the means for affecting lower centres by the products of the higher centres of the best men we had. We have already discovered that all advances in science and mathematics supply us with an unbelievable amount of purely psycho-logical and semantic data of extreme simplicity, which, without any technicalities, can be imparted to the masses in elementary structural education. Such education allows us to give very simply to children the 'cultural results', or to impart the *s.r*, which are the aim of university training, in a relatively short period and without any technicalities. These benefits, under an A education, are too rarely acquired even by university graduates, and impossible to impart to the masses, who are left helpless with archaic, delusional structural assumptions.

From one point of view the $\bar{A}$ issues are childishly simple and obvious, but from another, because of the power of old established habits and *s.r*, are quite difficult for the grown-ups to apply. It seems evident that an infant must be under the influence of the standards of evaluation of those who take care of him, automatically connected with the structure of the language he is taught. Under such unavoidable conditions, it is obvious that to give the full benefit of a $\bar{A}$-system in the training of children, parents and teachers should, themselves, have entirely absorbed these new standards.

A $\bar{A}$ civilization will require a unification of all existing human disciplines on the base of exact sciences. This unification will require all scientists, mathematicians, physicists, and psychiatrists included, to become acquainted and *fully* to practice $\bar{A}$ standards of evaluation. A $\bar{A}$ revision would have an international and interracial application, requiring a very thorough revision of all doctrines, a better acquaintance of

specialists in one field with the accomplishments in other fields, and an *up-to-date epistemology*. If we try to disregard epistemology consciously, we delude ourselves, as we cannot eliminate *some* epistemology as a foundation for our methods of evaluation, and, therefore, unconsciously retain some primitive epistemology which through inappropriate standards of evaluation, introduces semantic blockages.

Mach said long ago: 'Not every physicist is an epistemologist, and not every one must or can be. Special investigation claims a whole man, so also does the theory of knowledge.' The influence of Mach on modern science is well known; men such as the late Jacques Loeb, Einstein, the younger quantum pioneers., were deeply influenced by the writings of Mach, because Mach was a deep student of epistemology. But in a $\bar{A}$ society his statement must be slightly reworded; namely: 'Not every individual knows or realizes the importance of, or seemingly consciously cares for, epistemology; yet every one unconsciously has one and acts and lives by it. Each individual has his own special problems, the solution of which always claims the whole man, and no man is complete, unless he consciously realizes the permanent presence in his life of some standards of evaluation. Every one has thus *some* epistemology. There is no way of parting with it,—nor with air, nor with water,—and live. The only problem is whether his standards of evaluation are polluted with primitive remains of bygone ages, in a variety of ways; or sanified by science and modern epistemology.'

The present work shows how any system involves a special epistemology which we accept unconsciously, once we accept the system. To evaluate a system is practically equivalent to formulating its epistemology. This is strictly connected with linguistic and structural investigations.

To centralize and co-ordinate the $\bar{A}$ efforts, an *International Non-aristotelian Library* has been originated, which field embraces, ultimately, all known doctrines and human interests, the first publication being the present handbook. To facilitate the application of $\bar{A}$ disciplines and to stimulate further researches, an *International Non-aristotelian Society* has been incorporated with headquarters in New York City and branches to be established in all cities of the world which have educational institutions. The main aims of the Society are scientific and educational for the study, by means of papers and lectures followed by discussions, of the $\bar{A}$ aspects necessary for a revision and, therefore, a co-ordination of all existing sciences and concerns of man. As the aspects of science which are of interest to the Society would be *structural* and *semantic*, from the point of view of a *general theory of values*, the lectures would be of a general non-technical character on the level of intelligent laymen.

Science would not be 'popularized' but analysed from a fundamental $\bar{A}$ epistemological point of view, compelling the speakers and authors of papers to analyse the deeper *non-el,* structural, and semantic foundations of experience, as well as of theories. The layman would benefit because he would be given a structural education readily understood, without being led astray by the older A 'popularization'. Later, if economically feasible, it is intended to issue a monthly *International Non-aristotelian Review,* and also to organize *International Non-aristotelian Congresses.*

The A-system was the result of the *s.r* of the white race of more than two thousand years ago; it built up the doctrines, institutions . , appropriate to this system. In those days, knowledge was very scanty ; the interconnection of different peoples, vague; the means of communications, very primitive , . It may be considered that science, and particularly mathematics, began a $\bar{A}$ revolution by explicitly searching for structure and adjusting the structure of the scientific languages, which we usually call 'terminology', 'theories', . Modern conditions of life are, to a large extent, affected by $\bar{A}$ science but exploited by the thoroughly A doctrines of the commercialists, militarists, politicians, priests, lawyers . , which results in a bewildering chaos, resulting in needless, great, and imposed suffering for the great masses of mankind, as exemplified by such cataclysms as wars, revolutions, unemployment, different economic crises , .

$\bar{A}$ disciplines, or science *as such,* are thoroughly beneficial to mankind at large; but an A exploitation and use of these $\bar{A}$ products are, and must be, a source of endless sufferings to the enormous majority of mankind, leading automatically to every kind of break-down. It is impossible to give a fuller analysis of this complex interrelation, as this would require a separate volume; I shall, therefore, tabulate only a few overlapping suggestions.

Non-aristotelian, Scientific, Adult Standards of Evaluation	Aristotelian, Infantile Standards of Evaluation of Commercialism, Militarism , .
Biological Sciences.	
Particularly medicine has discovered means of how to keep or restore health, how to eliminate suffering, how to save and to prolong life . ; but medicine also gives means whereby we may prevent the overpopulation of this globe and so teaches us how to avoid great sufferings through overcrowding, which results in bitter struggle for food, shelter, employment . , in turn, leading to wars, revolutions, unemployment , .	Commercialized medicine unavailable to great masses of poor people. Commercialism, militarism, infantile dreams of 'world empires' sponsor unintelligent breeding in vast numbers which runs up the prices of land and houses, lowers the price of labour, and supplies cannon-fodder , . The intelligent control of numbers of population is prevented by jailing and otherwise persecuting scientific workers, which persecution affects only the poor and uneducated.

Non-aristotelian, Scientific, Adult Standards of Evaluation	Aristotelian, Infantile Standards of Evaluation of Commercialism, Militarism , .
Chemistry.	
Antiseptics, fundamental for medicine and the control of the increase or decrease in the population.	Sponsoring of overpopulation by forcibly withholding knowledge from the masses.
Drugs, fundamental for medicine.	Use of drugs in war to allow a soldier to suffer twice. Commercialized drug pedlars.
High explosives, necessary for agriculture, mining , .	High explosives used for killing in the struggle for ultimately futile 'world empires'.
Production of food products. Alcohol, wine, beer, .	Food destroyed to keep up prices. Commercialization of drinking. The saloon, crimes, 'prohibitions', the financing of gangsters, corruption of government and justice , .
Poisonous gas, necessary for the elimination of insects.	Poisonous gas in wars.
Linguistics.	
Law, as expressing some standards of evaluation.	Interpretations by commercialized lawyers to evade law, and the influencing of the wording of the law so as to make evasion possible. Lobbyists.
Newspapers, magazines . , giving most powerful educational means.	Commercialized newspapers . , controlled by profits and advertisements, supplying stultifying, controlled material, stimulating the morbid potentialities of the mob, to increase circulation , .
Other public prints, giving necessary or useful informations.	Commercialized advertisements. Schizophrenic play on words, the promoting of infantilism , .
Religions represent primitive structural rationalizations, or primitive 'science'; intended, also, as guides for conduct and adjustment, under the structural assumptions of the epoch of their primitive origin.	Commercialized religions. Religions having outlived their usefulness, often become priestcraft as a source of income and control. The sponsoring of primitive and delusional standards of evaluation often for private gains. The imposition of primitive standards of evaluation involves pathological factors.
Physics and Related Sciences.	
Aeroplanes, as means of communication and scientific exploring.	Militarized aeroplanes, as means of destruction and murder.
Automobiles, as means of transportation and pleasure.	Militarized automobiles, as means of destruction and suffering.

NON-ARISTOTELIAN, SCIENTIFIC, ADULT STANDARDS OF EVALUATION	ARISTOTELIAN, INFANTILE STANDARDS OF EVALUATION OF COMMERCIALISM, MILITARISM ,.
Physics and Related Sciences (continued).	
Machinery and tools, as means to eliminate avoidable efforts and to benefit fully through natural resources; to give greater comfort and sanitary conditions; to allow greater leisure for cultural pursuits ,.	Commercialized machines, as means for larger individual profits and suffering for the increasing numbers of unemployed and starving masses. Mass production of guns, ammunitions ., for mass extermination and destruction, and larger profits for the manufacturers and investors.
Moving pictures, as powerful educational means.	Commercialized moving pictures, stimulating and satisfying the erotic and morbid crowds, and for private propaganda.
Radio, as powerful means of communication and education.	Commercialized radio, advertisements, private propaganda, often stimulating morbid inclinations of the mob.
Railroads, as means of public transportation.	Commercialized railroads, as means for private gains and the control by a few of large areas and many people.
Tractors, as means of transportation and sources of power in agriculture.	Tractors, as tanks and means of destruction and murder in wars.
Public Servants.	
Judges, as guardians of some standards of evaluation.	Commercialized politicians, as judges, corruption, lack of justice ,.
Lawyers, as assistants in the administration of justice.	Commercialization and corruption of the legal profession. Means to evade or pervert justice.
Police, as an executive, regulating, safety-force.	Commercialization of police by politicians, corruption, combinations with the underworld ,.
Sports, as means for preserving and building up health, co-ordinated orientation, fair play, and in a lesser degree as recreation.	Commercialization of sports, elimination of benefits. Gambling. Lowering of educational standards ,.

From a $\bar{A}$ point of view, which eliminates primitive *s.r*, it becomes obvious that mankind represents an interdependent time-binding class of life, and any group of people who possess physical means for destruction and still preserve infantile standards of evaluation become a menace to the culture of the whole race. Under such conditions we must have agencies for an international exchange and evaluation of our standards, as well as methods which would help us in adjusting these standards.

At present, we must admit that with the modern, rapid, and international advancement of science we have fairly well-established interna-

tional $\bar{A}$ standards of scientific values. International Scientific Congresses are not only necessary for the advancement of science, but they also explicitly prove science to be entirely international.

The latest, most important $\bar{A}$ institution is found in the League of Nations, which embraces practically the whole civilized world, with the exception of a very few nations who display infantile and A aloofness, using different self-deceptive excuses, and, to a large extent, handicapping the power and usefulness of the League.

As we have learned lately, not only human achievements, but also human disasters, are mostly interrelated and international, and are becoming more so every year. Obviously, with A narrowness, selfishness, shortsightedness, infantilism, commercialism, militarism, nationalism., rampant, mankind, to prevent further major A disasters, would have to produce a special international body which would co-ordinate various structural achievements, strivings., formulate and inform the great masses of mankind of the modern scientific $\bar{A}$, adult standards of evaluation.

At present, we already have the necessary agencies; but, as yet, they are inefficient and non-co-ordinated. These are to be found in International Scientific Congresses, and the League of Nations. The weak spots of these organizations are found in the fact that the Scientific Congresses are too cumbersome, expensive, non-co-ordinated and only periodic. The League of Nations, although a $\bar{A}$ body in structure, is mostly made up of men who do not know any other standards of evaluation than A, and so they often lack the means to present a scientific, or $\bar{A}$ argument, and usually do not realize the tremendous power they would have in a $\bar{A}$-system. In human affairs, for instance, there cannot be a neutral and innocent absentee. One such absentee with guns and battleships becomes a powerful blocking and so, ultimately, disrupting factor for the rest of civilization. Such an absentee is not, then, guilty by omission; but, from a $\bar{A}$ point of view, becomes guilty by commission. The League, when definitely and fully allied with international science, will some day have the pluck to make such a declaration and act accordingly. A consciously $\bar{A}$ League of Nations will not limit itself to the thankless and very often useless task of adjusting inevitable clashes of A standards of evaluation, but will, with the full co-operation of scientists, undertake the much more important, constructive, and unique duty of a guardian and leader of human culture. Such a League would become a scientific, professional, international, co-ordinating, cultural, time-binding advisory organization for all nations. National A govern-

ments, instead of only *instructing* their *A* representatives, would first *consult* with the new *Ā specialists*.

Many politicians and their followers often become all but hysterical at the mention of the League of Nations, which, in some mysterious way, they associate with a 'super-state' or 'control',. Let me say, at once, that a symbolic, or human class of life, is very largely controlled by ignorant, hidden, often pathological., factors *beyond public control,* of which the majority are entirely innocent. In the human symbolic class of life no one is entirely free, but all our lives are entangled in an interdependence of human relations. The dependence on those powers which are now hidden, and *beyond public control,* constitutes a grave danger to all. Not so with a scientific, enlightened public opinion with adult standards of evaluation as formulated by a future co-ordination of science and the League of Nations. Such great majority-opinions will remain opinions, or statements of standards of evaluation, which any one member of the League may accept or reject; but, then, it would be necessary for him to state publicly his standards of evaluation and to decide consciously to act with or against, or to enlighten further the opinion of the human race. There is, of course, no question of 'super-states' or 'control' except the unified demand for a conscious and explicit stand on any important subject by any nation. Public opinion will do the rest, once it is convoked to act.

I am not a pacifist in the accepted sense. In an animalistic, infantile, or *A* society, this would be not only impossible but downright silly. Quite the contrary, I am disgusted with the infantile standards under which wars are conducted. Thus, our rulers and war lords, sponsored by commercialism, like little boys make a sort of game out of wars, and thus help to preserve them as an institution. In a consistent society, wars should be as ruthless as possible *to all.* If any one wants a war, he should consistently take all the consequences. But this would not suit our infantile rulers; they know that when little boys play at war, and one group becomes too rough, the other group refuses to play, and so the play at war ends.[3] All these perverted 'humanitarianisms' only sponsor wars because, in an unlimited modern warfare, the people would soon come to their senses and would refuse to suffer for the benefit of the very few. So I am far from being an *A* pacifist.

But why our destinies should be dependent on the accidental and primitive structure of the language we use., is beyond my comprehension. I grant that if we accept such and such postulates, two-valued, *el,* structurally false-to-fact *A* 'logics', 'psychologies'., all the old, too famil-

iar consequences follow, which we, in our ignorance, have forced upon human life.

But if we put all systems and all 'logics'., on new $\bar{A}$ foundations, which are structurally closer to the facts of life (1933), all the older conclusions may even be reversed. The problem now before mankind is whether or not the new $\bar{A}$-system is more similar in structure to the world and our nervous system than the old. On the answer to this question, the future of civilization depends.

From the present point of view, we should establish with the League of Nations a permanent $\bar{A}$ or scientific department, composed of a few of the best scholars from all countries, who would keep in touch, not only with the developments of their specialties, but would also co-ordinate them on general structural and epistemological foundations. This department would be the international authority on modern revised and co-ordinated standards of evaluation, which would be published in special proceedings. The present discrepancy and lack of co-ordination between different branches of knowledge becomes genuinely alarming and detrimental to mankind, because in 1933 it is humanly impossible for a single individual to attempt such a co-ordination. Members of this group would be selected by the universities of each country. In their researches, joint studies, and results, mankind at large would find the most reliable scientific and $\bar{A}$ opinions produced at each date, and would have some definite and conscious standards of evaluation by which to orient themselves.

The modern 'voting' has some benefits in local affairs, but when its very limited validity is not understood, it becomes a serious danger to mankind. Thus, when we are ill, or when we want a bridge built, we ask specialists for their scientific co-operation; we would hardly depend on ignorant voters. Similarly, in a $\bar{A}$ scientific civilization, the major problems of mankind would be analysed by scientific specialists, recommendations offered to be accepted or rejected, as the case may be; but the ignorant voter would have at his disposal an unbiased, impersonal, and responsible opinion of international scientific specialists to compare with the equivocations of some local ignorant politician.

To facilitate such future $\bar{A}$ activities, the International Non-aristotelian Society has been established. It is hoped that soon the scientific, educational, 'mental' hygiene., workers will begin to unite on a local and national $\bar{A}$ basis. Later, International Congresses will unite the local societies, which ultimately will be embodied as a permanent institution, most probably in the League of Nations.

In the process of formulating the above system a curious observation has been forced upon me; namely, that statements which are, for instance, quite legitimate for the English language, even though they probably apply in general to all Indo-european languages, do not apply in a similar degree.

I am intimately acquainted with six languages, two Slavic, two Latin, and two Teutonic, and also with the psycho-logical trends of these groups. I have been led to suspect strongly that the finer differences in the structure of these languages and their use are connected with the semantics of these national groups. An enquiry into this problem, in my opinion, presents great semantic possibilities and might be the foundation for the understanding of international psycho-logical differences. Once formulated, this would lead us to a better mutual understanding, particularly if a $\bar{A}$ semantic revision of these different languages is undertaken. To the best of my knowledge, this field of enquiry is entirely new and very promising.

It must be obvious to the reader that such a vast program is beyond the power of a single man to carry out, and the present author hopes for public interest in this enterprise.

If the $\bar{A}$-system has accomplished nothing more than to draw the attention of mankind to some disregarded problems; if it has done nothing more than point the way, not to panaceas, but to suggestions toward an expedient, constructive, and unified scientific program whereby future disasters may be avoided or lessened—the writer will be satisfied.

BOOK III

ADDITIONAL STRUCTURAL DATA ABOUT LANGUAGES AND THE EMPIRICAL WORLD

The every-day language reeks with philosophies . . . It shatters at every touch of advancing knowledge. At its heart lies paradox.

The language of mathematics, on the contrary, stands and grows in firmness. It gives service to men beyond all other language. (25)

ARTHUR F. BENTLEY

Nothing is more interesting to the true theorist than a fact which directly contradicts a theory generally accepted up to that time, for this is his particular work. (415)

M. PLANCK

It is not surprising that our language should be incapable of describing the processes occurring within the atoms, for, as has been remarked, it was invented to describe the experiences of daily life, and these consist only of processes involving exceedingly large numbers of atoms. Furthermore, it is very difficult to modify our language so that it will be able to describe these atomic processes, for words can only describe things of which we can form mental pictures, and this ability, too, is a result of daily experience. (215)

W. HEISENBERG

PREFATORY REMARKS

In re mathematica ars proponendi quaestionem pluris facienda est quam solvendi. (74)
GEORG CANTOR

We cannot describe substance; we can only give a name to it. Any attempt to do more than give a name leads at once to an attribution of structure. But structure can be described to some extent; and when reduced to ultimate terms it appears to resolve itself into a complex of relations . . . A law of nature resolves itself into a constant relation, . . . , of the two world-conditions to which the different classes of observed quantities forming the two sides of the equation are traceable. Such a constant relation independent of measure-code is only to be expressed by a tensor equation. (148)
A. S. EDDINGTON

We have found reason to believe that this creative action of the mind follows closely the mathematical process of Hamiltonian differentiation of an invariant. (148)
A. S. EDDINGTON

The only justification for our concepts and system of concepts is that they serve to represent the complex of our experiences; beyond this they have no legitimacy. I am convinced that the philosophers have had a harmful effect upon the progress of scientific thinking in removing certain fundamental concepts from the domain of empiricism, where they are under our control, to the intangible heights of the *a priori*. (152)
A. EINSTEIN

In writing the following *semantic* survey of a rather wide field of mathematics and physics, I was confronted with a difficult task of selecting source-books. *Any* mathematical treatise involves conscious and many unconscious notions concerning 'infinity', the nature of numbers, mathematics, 'proof', 'rigour'. , which underlie the definitions of further fundamental terms, such as 'continuity', 'limits', . It seems that when we discover a universally *constant empirical relation*, such as 'non-identity', and apply it; then all other assumptions have to be revised, from this new point of view, irrespective of what startling consequences may follow.

At present, neither the laymen nor the majority of scientists realize that human mathematical behaviour has many aspects which should never be identified. Thus, (1) to be somehow aware that 'one and one combine in some way into two', is a notion which is common even among children, 'mentally' deficients, and most primitive peoples. (2) The mathematical '1+1=2' already represents a very advanced stage (in theory, and in method . ,) of development, although in *practice* both of these *s.r* may lead to one result. It should be noticed that the above (1) represents an individual *s.r*, as it is not a general formulation; and (2) represents and involves a generalized *s.r*. Does that exhaust the problem of '1+1=2'? It does not seem to. Thus, (3), in the *Principia Mathematica* of Whitehead and Russell which deals with the meanings and foundations of mathematics, written in a special shorthand, abbreviating statements perhaps tenfold, it takes more than 350 large 'shorthand' pages to arrive at the notion of 'number one'.

It becomes obvious that we should not identify the manipulation of mathematical symbols with the semantic aspects of mathematics. History and investigations show that both aspects are necessary and important, although of the two, the semantic discoveries are strictly connected with the revolutionary advances in science, and have invariably marked a new period of human development. In Chapter XXXIX, the reader will find a very impressive example of this *general* fact. Thus, what is known as the 'Lorentz transformation', *looks like* the 'Einstein transformation'. When manipulated numerically both give equal numerical results, yet the meanings, and the semantic aspects, are different. Although Lorentz produced the 'Lorentz transformation' he did not, and *could not* have produced the revolutionary Einstein theory.

It is well known that when it comes to the manipulation of symbols mathematicians agree, but when it comes to the semantic aspects or meanings . , they are *admittedly* hopelessly at variance. In a prevailingly A world we have had no satisfactory theory of 'infinity', or a $\bar{A}$ definition of numbers and mathematics. This necessarily resulted in the fact that the semantic aspects of practically all important mathematical works by different authors often involve *individual semantic presuppositions*, or orientations concerning fundamentals. My presentation intends to be primarily semantic and elementary, and is only remotely concerned with the manipulation of symbols. A $\bar{A}$-system, which rejects 'identity', differs very widely from A attitudes, and introduces distinct $\bar{A}$ requirements. I had, therefore, to select from many works, with their *individual* presuppositions, those which were less in conflict with $\bar{A}$ principles than the others.

A survey of important mathematical treatises shows that although the majority of modern mathematicians explicitly abjure the 'infinitesimal', yet, in some presentations, this notion persists. In my presentation I reject the 'infinitesimal' explicitly and implicitly, although the formulae are not altered. 'Modern' calculus is based *officially* on the theory of limits, but as the theory of limits involves the unclarified theory of 'infinity'. , nothing would be gained semantically and for my purpose, had I stressed these formal possibilities of the calculus. Quite the opposite, if I had done so, I would have failed to stress the most fundamental $\bar{A}$ principle and task of establishing the *similarity of structure between languages and the un-speakable levels and happenings as the first and crucial consequence of the elimination of identity.* For these weighty reasons, in my presentation, I followed some older textbooks, particularly Osgood's, which, from a $\bar{A}$ point of view, are sounder than the newer, largely A rationalizations and apologetics.

However, it should be realized that practically all outstanding and creative mathematicians have had, and still have, $\bar{A}$ attitudes, yet, these private beneficial attitudes, not being formulated in a $\bar{A}$-system, could not become *conscious*, simple, workable, public, and educational assets. We can be simple about this point. With the elimination of identity, structure becomes the *only possible* content of 'knowledge'—and structure of the un-speakable levels has to be *discovered.* Discovery depends on the finding of *new*, and therefore *different*

characteristics. In the *formulation* of the last sentence, we cannot make the 'training in discovery' an *educational discipline*. The opposite is true in a $\bar{A}$-system, based on non-identity, as *we can train simply and effectively in non-identity*, which ultimately leads to differentiation, and so discovery.

Because of the elementary, and purely semantic character of the following pages, I have often restrained myself from giving technical, supposedly 'rigorous', and often A rationalizations, which we occasionally call definitions. In a semantic and $\bar{A}$ treatment, at this pioneering stage, stressing old definitions would be seriously confusing; and I wished to avoid such witty wittgensteinian 'definitions' as 'A point in space is a place for an argument'. In a number of instances, and for my purpose, I often avoided unsatisfactory formal definitions, preferring to depend upon the ordinary meanings of words.

For the reader who wishes to acquaint himself with an elementary theory of limits and corresponding sets of definitions, I would suggest the book of the late Professor J. G. Leathem, *Elements of the Mathematical Theory of Limits* (London and Chicago, 1925). This theory is based on Pascal's *Calcolo Infinitesimale*, Borel's *Théorie des fonctions*, and Godefroy's *Théorie des séries*. Leathem's book has been printed under the supervision of Professor H. F. Baker, F.R.S., of the University of Cambridge, and Professor E. T. Whittaker, F.R.S., of Edinburgh. I give these names for professional mathematicians, to indicate the semantic trend which underlies this particular treatment of limits and which does not greatly conflict with a $\bar{A}$ outlook. This outlook may be summarized in part, in the words of Borel somewhat as follows: 'To the evolution of physics should correspond an evolution of mathematics, which, without abandoning the classical and well-tried theories, should develop however, with the results of experiments in view'. This statement implies vaguely the 'similarity of structure'. , and so requires as a *modus operandi* the rejection of identity.

There seems to be little doubt that a complete and radical revision of the semantic aspects of human mathematical behaviour is pending. Such a revision appears to be laborious and difficult, and should be undertaken from the point of view of the theory of the unique and specific relations, called numbers. I doubt if a single man could accomplish this revision. Such an undertaking will probably be the result of group activities, and may, in the beginning, be *unified* by the formulation of one fundamental $\bar{A}$ principle of non-identity, the disregard of which, from science down to 'mental' ills, can be found at the bottom of practically all avoidable human difficulties.

The problems are very complicated and extremely difficult, and need to be treated from many angles. At present, we have many scientific societies, grouped by their specialties; but we do not have a scientific society composed of *many different specialists* whose work could be unified by some *common and general principle*. There can be no doubt that the principle of 'identity', or 'absolute sameness in *all* aspects', is invariably false to facts. The main problem is *to trace* this semantic disturbance of improper evaluation in all fields of science and life, and this requires a new *co-ordinating scientific body* of many specialists, with branches in all universities. Each group would meet, say monthly, to

discuss their problems, and give mutual technical assistance in tracing *this first general semantic disturbance.* Such meetings would stimulate enormously scientific productivity. In fact, without such a co-ordinating body, the present enormous technical developments in each branch of science preclude the revision of general principles, on which, in the last analysis, all other of our activities greatly depend. The first task then, is to find a co-ordinating principle, and present it to the scientific world.

Psychiatry, and common experience, teach us, that in heavy cases of dementia praecox we find the most highly developed 'identification'. $\bar{A}$ considerations suggest that *any* identification, no matter how slight, represents a dementia praecox factor in our semantic reactions. The rest is only a question of degrees of this maladjustment. From this point of view, we will find dementia praecox factors even in mathematics. In physics, only since Einstein has this factor of un-sanity been eliminated, and this elimination has already produced an ever-growing crop of 'geniuses', which merely means, that some inhibitions of mis-evaluation have been eliminated from these younger men, and that they are humanly more 'normal' than the others.

In mathematics, from a $\bar{A}$ point of view, we must first of all *not identify* different aspects of our mathematical behaviour, nor try to cover up these identifications of endless aspects by the one very old term 'mathematics'. This word, 'mathematics', in its *accepted sense* covers a non-existing fiction. What does exist, and the only thing we actually deal with, is *human mathematical behaviour*, human *s.r*, and the *results* of human *mathematical behaviour* and *s.r.* A treatise, say, on a new quantum mechanics, has no value to a monkey or a corpse, and only human *mathematical behaviour* and *s.r*, have any actual *non-el* existence, and is the *only* thing which actually matters. So we see that 'mathematics' covers a non-existent fiction *if elementalistically* separated from human mathematical behaviour and *s.r.* I use the term 'mathematics' in the *non-el* sense, and attempt to signalize some of the difficulties non-elementalism involves at this transitory stage.

From a $\bar{A}$, non-identity, structural, *non-el* point of view, human mathematical behaviour must be treated uniquely as a physico-mathematical discipline, and postulational methods to be used exclusively as a most valuable *checking* method. To *base* mathematical behaviour and *s.r* on postulational methods exclusively, is to introduce dementia praecox factors into science, which only induces the spread of semantic maladjustment in life.

Our main task in producing a $\bar{A}$ revision of mathematical *s.r*, is in the elimination of identification from our *s.r* about 'infinity' and in the formulation of a $\bar{A}$ definition of numbers in terms of relations. This would enable us to rebuild human mathematical *s.r* from a theory of numbers point of view, as a *physico-mathematical* discipline. The intrinsic, or internal theory of surfaces, and the tensor, or absolute calculus, are methodologically our most secure epistemological guides.

I would suggest that mathematical and scientific readers who are interested in a $\bar{A}$ revision should, at first, in their special fields, sketch in technical papers,

presented before the local International Non-aristotelian Societies, A pitfalls and $\bar{A}$ problems and outlooks. Only after this is done, shall we be able to begin a co-ordination of their findings, and thereby initiate a revised and unified $\bar{A}$ science, mathematics, and perhaps ultimately a saner *scientific civilization.*

The scientific achievements dealt with in Book III, are developing so rapidly, and the technical points of view alter so often, that on a static printed page it is impossible to do them justice. The writer has spared no efforts to keep informed of these scientific developments until two weeks before the appearance of this book; yet because these new developments do not represent new and fundamental semantic factors, I deliberately do not include them here. In some instances, a given author may seem to change his opinions, but, from a $\bar{A}$ point of view, it sometimes appears that the original notions were more justified, and so I preserved them without change.

The following pages are written exclusively from a semantic point of view, an undertaking which is far more difficult than dealing with a restricted technical physico-mathematical problem, because it involves *second order* observations, of the first order observations, of the first order observer, and of the relations between them , . When it came to a final revision of the manuscript, and reading of the proofs, I found that dealing with so many varied fields, languages, and symbolisms at one period, was no small task, and I only hope that I have not over-looked too many errors or misprints.

If we must have slogans, a $\bar{A}$ motto readily suggests itself—'Scientists of the world unite'. Perhaps this motto may prove more constructive and workable than the familiar A *elementalistic* slogans which have mostly led to the dismembering of human society. Protests against any misrule should not be confused with the proclaiming of disrupting *general principles.* Let me repeat once more, that the most lowly manual worker is useful *only* because of his human nervous system, which produced all science, and which differentiates him from an animal, and not primarily for his hands alone; otherwise we would breed apes to do the world's work.

In the explanations of some geometrical notions, and some parts of the theory of Einstein, I have followed often very closely the *Einstein's Theory of Relativity* by Max Born, which is easily the best elementary exposition I have read, and also the books of Eddington. In the quantum field I have followed mostly the books by Biggs, Birtwistle, Bôcher, Haas, and Sommerfeld, and I wish to acknowledge my indebtedness to the above authors.

I am also under heavy obligations to Professors E. T. Bell, P. W. Bridgman, B. F. Dostal, R. J. Kennedy, and G. Y. Rainich, who were so kind as to read the MS. and/or proofs, and whose criticism and suggestions were invaluable to me. However, I assume entire responsibility for the following pages, especially since I have not always followed the suggestions made.

PART VIII

ON THE STRUCTURE OF MATHEMATICS

Being myself a remarkably stupid fellow, I have had to unteach myself the difficulties, and now beg to present to my fellow fools the parts that are not hard. Master these thoroughly, and the rest will follow. What one fool can do, another can. (510) SILVANUS P. THOMPSON

Besides the theory of surfaces is the model on which all the higher theories are built and must be built, and it is well to master it completely before attempting generalizations. (425) G. Y. RAINICH

To find such relations Einstein has applied a mathematical method of great power—the calculus of tensors—with extraordinary success. The calculus threshes out the laws of nature, separating the observer's eccentricities from what is independent of him, with the superb efficiency of a modern harvester. (21) E. T. BELL

CHAPTER XXXII

ON THE SEMANTICS OF THE DIFFERENTIAL CALCULUS

The principle of gaining knowledge of the external world from the behaviour of its infinitesimal parts is the mainspring of the theory of knowledge in infinitesimal physics as in Riemann's geometry, and, indeed, the mainspring of all the eminent work of Riemann, in particular, that dealing with the theory of complex function. (547) HERMANN WEYL

The conception of tensors is possible owing to the circumstance that the transition from one co-ordinate system to another expresses itself as a **linear** transformation in the differentials. One here uses the exceedingly fruitful mathematical device of making a problem "linear" by reverting to infinitely small quantities. (547) HERMANN WEYL

Section A. Introductory.

In the first draft of this book written in 1928, the following pages preceded Part VII. In a final revision in 1932, it seemed advisable to transfer pages which to laymen look 'mathematical', to the end of the volume, because the majority of even intelligent readers have a sort of 'inferiority complex' about anything 'mathematical'.

The patient reader knows by now, I hope, that on neurological grounds, he must for the sake of sanity, be able to translate the dynamic into the static, and the static into the dynamic; and also that he must know at least about the modern structure of 'space', 'time', 'matter'. These conditions seem essential for sanity, and so I had no choice but to give the minimum of a structural and semantic outline, and to acquaint the reader with the existence of modern scientific problems and vocabularies. It is not my aim to teach the reader mathematics or modern physics. I must limit myself to structural and semantic issues, for there are excellent elementary books which will give him the necessary informations.

The following pages should in no way intimidate the intelligent reader. Elementary structural statements and definitions are given in simple language, followed by illustrations to render their meanings more understandable. The pages are less technical than they look, as each example is carried through in the most elementary way in all of its details, so as to make easy reading. A real difficulty for some readers may come from the semantic blockage created by the use of apparently strange, and, to them, unknown terms, or from a feeling of fright or abhorrence of anything mathematical, due to deplorably faulty introduction to some branch of mathematics at the hands of some teacher innocent of the broader epistemological aspects of science. I am acquainted with scientists of very considerable mathematical gifts, who have had to overcome this phobia of mathematics. Once the word 'mathematics' was mentioned to them, they became 'mentally' paralysed. An 'emotional' fright seized them and it took some months to overcome this undesirable childish *s.r.* I use the subject of mathematics as an illustration of this difficulty, because I want to contrast the comparative simplicity of mathematical notions

with the complexity of human problems and language. For when we have understood the *simplest* notions, which happen to be mathematical, then only shall we be able better to understand our human problems, which are in comparison so difficult and so confused.

Any reader who has a distaste for mathematics will benefit most if he overcomes his semantic phobia and struggles through these pages, even several times. As a result of so doing he will find it simple although not always easy. It is always semantically useful to overcome one's phobias; it liberates one from unjustified fears, feelings of inferiority , . The main point of this whole discussion is to evoke the semantic components of a living Smith, when he habitually uses the method which will be explained herewith. This method is so simple and so fundamental that in the form given by a $\bar{A}$-system and further simplified according to the gifts of the teacher, it will some day be introduced into *elementary* schools without technicalities, as a *preventive* semantic method against 'insanity', un-sanity and other nervous and semantic difficulties, and as a foundation for a training in *sanity* and adjustment.

Section B. On the Differential Calculus.

1. *GENERAL CONSIDERATIONS*

As we have already seen, the structural notion of a function is strictly connected with that of the variable. The variable on one level does not 'vary'; it is a selection by Smith of a definite value from a given set. As these processes are going on inside of the skin of Smith he might experience on a different level a feeling of 'change'. The method of dealing with such problems is given by the mathematical differential and integral calculus.

The beginnings of methods dealing with 'change' are to be found even among the ancients. Galileo, Roberval, Napier, Barrow, and others were interested in 'fluxional' methods, before Newton and Leibnitz.[1] The epoch-making discoveries of the last two mathematicians consisted not only in perfecting the knowledge they had and in inventing new methods, but also—and this is perhaps the most important—they formulated a *general* theory of these structural methods and invented a new notation suitable for their purpose. The definite abandonment of the old tentative methods of integration in favour of methods in which integration is regarded as the inverse of differentiation was especially the work of Newton. Leibnitz' main work was in the field of precise formulation of simple rules for differentiation in special cases and the introduction of a very useful notation.

It is not an exaggeration to say that the calculus is one of the most inspiring, creative, structural methods in mathematics. There is little doubt that the analysis of the foundations of mathematics, and their revision, was suggested by a study of the methods of the calculus. *It is structurally and semantically the 'logic' of sanity* and, as such, can be given ultimately without technicalities by the present $\bar{A}$-system and semantic training, with the aid of the Structural Differential.

The application of the differential calculus to geometry produced differential geometry. This prepared the way for the notions of Einstein and Minkowski.

The whole of modern physics becomes possible through the calculus, and it will probably be correct to say that the achievements of the future also will be dependent on it.

The present work is also to a large extent inspired by it, and develops simple non-technical methods by which the psycho-logical structural *s.r* necessitated by the calculus can be given to the masses in elementary education without any technical knowledge of it. This statement does not include teachers, who should be acquainted with at least the rudiments of the calculus.[2]

It is true that in the beginning we did not suspect that the semantics of the calculus are indispensable in education for *sanity*. It is the *only* structural method which can reconcile the as yet irreconcilable higher and lower order abstractions. Without such a reconciliation, at our present level of development, sanity is a matter of good luck quite beyond our conscious or educational control.

Let us recall the rough definition of a function: y is said to be a function of x if, when x is given, y is determined. In symbols we write $y=f(x)$ which we read 'y is equal to a function of x' or 'y is equal to f of x'. If y is a function of x, or $y=f(x)$, then x is called the independent variable, being the one to which we arbitrarily assign *any* value we choose out of a given set of values. The y is called the dependent variable as its value depends on the value we assign to x.

A function may have more than one independent variable; in which case we have a function of several variables. It happens frequently that to one value of the independent variable there may correspond several values of the dependent variable. Then y is said to be a multiple-valued function of x.

Roughly speaking, a function is said to be continuous if a small increment in the variable gives rise to a small increment of the function.

A theory of functions can be developed without any references to graphs and geometrical notions of co-ordinates and lengths; but in practice (and in this work), it is extremely useful to introduce these geometrical notions, as they help intuition. A modern definition of an analytic function is technical and unnecessary for our purpose. Suffice it to say that it is connected with derivatives and power series, *which means structure.*

Geometry is a very remarkable science. It may be treated as pure mathematics, or it may be treated as physics. It may therefore be used as a link between the two or as a link between the higher and lower order of abstractions. This fact is of tremendous psycho-logical and semantic importance. It is not by pure 'chance' that the most important writers on mathematical philosophy, authors who have generalized their knowledge of mathematics to include human results, were mostly geometers.

Indeed, Whitehead, in his *Universal Algebra* (p. 32), says, and justly so, that a treatise on universal algebra is also a treatise on certain generalized notions of 'space'. 'Space' should be understood as 'fulness', 'fulness of some-

thing', a plenum. Naturally coherent speech, like universal algebra, must be coherent speech about *something*. 'Generalized space' becomes generalized plenum, and so it belongs to *two* realms. One is contentless and formal, hence generalized algebra; the other, in that it refers to a generalized plenum, becomes generalized geometry, or generalized physics.

The main importance, perhaps, of geometry is in the fact that it can be interpreted *both ways*. One way appears as pure mathematics, and therefore as the study of sets of numbers representing co-ordinates. The other takes the form of an interpretation, in which its terms imply a connection with the empirical entities of our world. Obviously if speech is not the things spoken about, we must have a special discipline which will translate the coherent language of pure mathematics, which is contentless by definition, into another way of speaking which uses a different vocabulary capable of *both* interpretations.

Again, the different orders of abstractions, which our nervous structure produces, are perfectly reflected in the very structure and methods of mathematics. The possibility of the use of the 'intuitions' of lower order abstractions, is extremely useful in pure mathematics. This fact makes geometry also *unique*. It allows us to apply to the development of geometry both orders of abstractions—the 'intuitions', 'feelings', of the lower order of abstractions, and the static, 'quantum' jump methods of pure analysis. This is also why the einsteinian physics becomes four-dimensional geometry; which, because it can be treated on both levels of abstraction, gives tremendously powerful and important psycho-logical means for sanity and nervous co-ordination of the individual. Since Einstein, many far-sighted scientists have said that although they do not know in what respect the Einstein theory will affect our lives, yet they feel that it will have a tremendous influence. I venture to suggest that the bearing of the Einstein theory and its development on the problems of sanity, as explained in this work, is a new and unexpected semantic result of the application of modern science to our lives. As the Einstein theory could have been formulated more than two hundred years ago when the finite velocity of light was discovered, so the present theory is also several hundred years overdue. The only consolation we have left is that it is better late than never.

The scope of this work allows us to go but a little beyond these simple remarks, and permits only a very brief explanation of the most fundamental and elementary beginnings of the calculus. In this presentation I shall appeal very often to intuition (lower order abstractions), as this will help the reader.

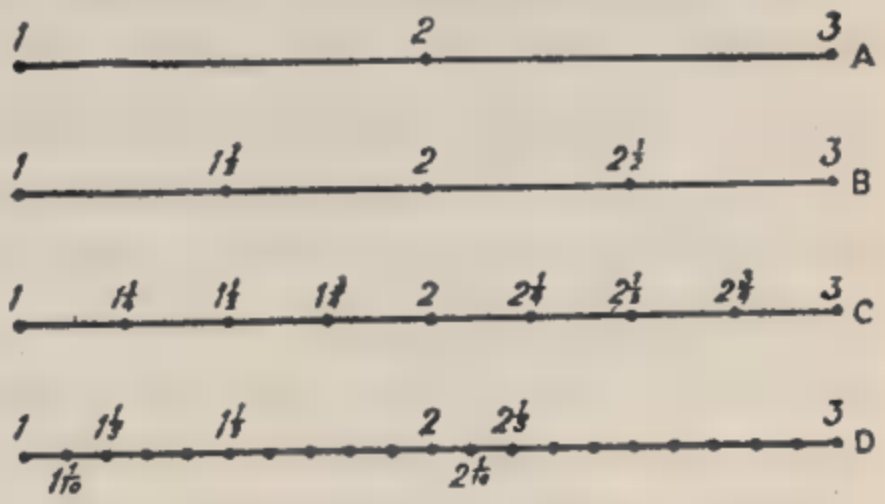
The notion of differentiation of a continuous function is the process for measuring the rate of growth; that is to say, the evaluation of the increment of the function as compared with the growth or increment of the variable. We may describe this process as follows: If y is a function of x, it is helpful not to consider x as having one or another special value but as flowing or growing, just as we feel 'time' or follow the ripples made by a stone thrown into a pond.

The function y varies with x, sometimes increasing, sometimes decreasing. We have already defined the variable as *any* value selected from a given range. Let us consider our x as given in the interval between 1 and 5. We are now

interested in all values which our x may take between these two values, or, as we say, in this interval. Obviously, we can select a few values, or, in other words, take big steps; as, for instance, assigning to x the successive values $x_1=1$, $x_2=2$, $x_3=3$, $x_4=4$, $x_5=5$. In such a case we would have few values and the difference between two successive values would be rather large, for instance, $x_3-x_2=1$. But such large differences are not of much interest to us here. We may, if we choose, select smaller differences; in other words, assign more values to our variable in the given range.

Let us take, for instance, for our x the series of values: 1, $1\frac{1}{2}$, 2, $2\frac{1}{2}$, 3, $3\frac{1}{2}$, 4, $4\frac{1}{2}$, 5. Here we see that the difference between two successive values is smaller than 1, it is $\frac{1}{2}$. So we already have nine, instead of the former five, values which we may assign to our x. Thus we have selected smaller steps by which to proceed. Let us select still smaller steps; for instance, $\frac{1}{4}$. Our extensional manifold of values for x in the interval between 1 and 5 would then be: 1, $1\frac{1}{4}$, $1\frac{1}{2}$, $1\frac{3}{4}$, 2, $2\frac{1}{4}$, $2\frac{1}{2}$, $2\frac{3}{4}$, 3, $3\frac{1}{4}$, $3\frac{1}{2}$, $3\frac{3}{4}$, 4, $4\frac{1}{4}$, $4\frac{1}{2}$, $4\frac{3}{4}$, 5. We see that in the interval between 1 and 5, we have already 17 values which we may assign to our variable, but we have followed the 'growth' of our x by smaller steps; namely, by steps of $\frac{1}{4}$. If we choose to diminish the steps to $\frac{1}{10}$, we would have for our extensional manifold of values: 1, 1.1, . . . , 1.9, 2, 2.1, . . . , 2.9, 3, 3.1, . . . , 3.9, 4, 4.1, . . . , 4.9, 5: in all, 41 values for x, any two succeeding values differing by $\frac{1}{10}$. If we select still smaller steps—let us say, $\frac{1}{100}$—we have 401 values for x and the difference between two successive values is still smaller; namely, $\frac{1}{100}$. This process may be carried on until we have as many numbers between 1 and 5 as we choose, since we may make the difference between successive numbers in the sequence as small as we please. In the limit, between any two numbers, let us say, 1 and 2, or any two fractions, there are infinite numbers of other numbers or fractions. It is obvious that in a given interval, let us say, between 1 and 5, we can have an indefinitely large number of intermediary numbers arranged in an increasing progression, such that the difference between two successive numbers can be made smaller than any assigned value, which is itself greater than zero.

The above may be made clearer by a geometrical illustration. Let us take a segment of a line of definite length, let us say 2 inches. Let us designate the ends by numbers 1 and 3. In figure (A) we divide the segment into 2 equal parts of one inch each, and see that to reach 3 starting with 1 we have to proceed by two large jumps from 1 to 2, and from 2 to 3. In figure (B) we have more steps in the interval and therefore the steps are smaller. In figures (C) and (D) the steps are still smaller and their number greater. If the number of steps is very large, the steps are very small. In the limit, if the numbers of steps become infinite, the length of the steps tends



toward zero and the aggregate of such points of division represents (in the rough only) a *continuous* line.

It is important that the reader should become thoroughly acquainted with the above simple considerations as they will be very useful in *any* line of endeavour. Here we already have learned how, somehow, to translate discontinuous jumps into 'continuous' smooth entities. Because of the structure of our nervous system we 'feel' 'continuity', yet we can analyse it into a smaller or larger number of definite jumps, according to our needs. The secret of this process lies in assigning an increasing number of jumps, which as they become vanishingly small, or tend to zero, as we say, cease to be felt as jumps and are felt as a 'continuous' motion, or change, or growth or anything of this sort.

An excellent example is given by the motion pictures. When we look at them we see a very good representation of life with all its continuity of transitions between joy and sorrow. If we look at an arrested film we find a definite number of *static* pictures, each differing from the next by a measurable difference or jump, and the joy or sorrow which moved us so in the play of the actors on the *moving* film, becomes a static manifold of static pictures each differing measurably from its neighbour by a slightly more or less accentuated grimace. If we increase the number of pictures in a unit of 'time' by using a faster camera and then release this film at the ordinary speed, we get what is called slow motion pictures with which we are all familiar. In them we notice a much greater smoothness of movements which in life are jerky, as, for instance, the movements of a running horse. They appear smooth and non-jerky, the horse looks as if it were swimming. Indeed we do swim no less than fishes, except that our medium; namely, air, is less dense than water, and so our movements have to be more energetic to overcome gravitation. The above example is indeed the best analogy in existence of the working of our nervous system and of the difference between orders of abstractions. Let us imagine that some one wants to *study* some event as presented by the moving picture camera. What would he do? He would first see the picture, in its moving, dynamic form, and later he would arrest the movement and devote himself to the contemplation of the static extensional manifold, or series, of the static pictures of the film. It should be noticed that the differences between the static pictures are finite, definite and *measurable.*

The power of analysis which we humans possess in our higher order abstractions is due precisely to the fact that they are *static* and so we can take our 'time' to investigate, analyse , . The lower order abstractions, such as our *looking* at the moving picture, are shifting and non-permanent and thus evade any serious analysis. On the level of *looking* at the *moving* film, we get a general *feeling* of the events, with a very imperfect memory of what we have seen, coloured to a large extent by our moods and other 'emotional' or organic states. We are on the shifting level of lower order abstractions, 'feelings', 'motions', and 'emotions'. The first lower centres do the best they can in a given case but the value of their results is highly doubtful, as they are not especially reliable. Now the higher order abstractions are produced by the

higher centres, further removed, and not in direct contact with the world around us. With the finite velocity of nerve currents it takes 'time' for impulses to reach these centres, as the cortical pathways offer higher neural resistances than the other pathways.[3] So there has to be a survival mechanism in the production of nervous means for arresting the stream of events and producing *static* pictures of permanent character, which may allow us to investigate, verify, analyse , . It must be noticed that because of this higher neural resistance of higher centres and the static character of the higher abstractions, these abstractions are less distorted by affective moods. For, since the higher abstractions persist, if we care to remember them, and the moods vary, we can contemplate the abstractions under different moods and so come to some *average* outlook on a given problem. It is true that we seldom do this, but we *may* do it, and this is of importance to us.

As one of the aims of the calculus is to study relative rates of change we will consider a series of successive values of our variable which differ by little from each other. If we have $y=f(x)$ we can consider the change in x for a short interval, let us say, from x_0 to x_1, so that we assign to our x two values, $x=x_0$ and $x=x_1$. The corresponding values of our function or y will be $y_0=f(x_0)$ and $y_1=f(x_1)$. In general, small changes in y will be almost proportional to the corresponding changes in x, provided $f(x)$ is 'continuous'.

Denoting the small increment of x byΔx, so that $x_1-x_0=\Delta x$ or $x_1=x_0+\Delta x$, function y receives the increment $y_1-y_0=\Delta y$ or $y_1=y_0+\Delta y$. Since $y_1=f(x_1)$ and $x_1=x_0+\Delta x$ we have:

$$y_0+\Delta y=f(x_0+\Delta x);$$ if we subtract

from both sides $$y_0=f(x_0)$$ we would have

$$\Delta y=f(x_0+\Delta x)-f(x_0);$$ dividing both sides

by Δx we have
$$\frac{\Delta y}{\Delta x}=\frac{f(x_0+\Delta x)-f(x_0)}{\Delta x} \tag{1}$$

The above ratio represents the ratio of the increment of the function to the increment of the variable. In the limit when the increment in the variable becomes vanishingly small or when Δx tends toward zero, and our function is continuous, the limit of this ratio gives us the law of change or growth of our function.

The limit which the ratio (1) approaches when Δx approaches 0,

$$\lim_{\Delta x\to 0}\frac{\Delta y}{\Delta x}=\lim_{\Delta x\to 0}\frac{f(x_0+\Delta x)-f(x_0)}{\Delta x} \tag{2}$$

is called the derivative of y with respect to x and is denoted by $D_x y$, which we read 'D_x of y', in symbols,

$$\lim_{\Delta x\to 0}\frac{\Delta y}{\Delta x}=D_x y \tag{3}$$

Let us illustrate this by a simple numerical example. Take the equation $y=x^2$ and assume that $x=100$, whence $y=10{,}000$. Suppose the increment of x, namely, $\Delta x=1/10$. Then $x+\Delta x=100.1$ and $(x+\Delta x)^2=100.1\times 100.1=10020.01$. The last 1 is 1/100 and only one millionth part of the 10,000, and so, we can

neglect it and consider $y+\Delta y=10{,}020$; whence $\Delta y=20$ and $\Delta y/\Delta x=20/0.1=200$. In the general case, if $y=x^2$ and instead of x we take a slightly larger value, $x+\Delta x$, then our function y also becomes slightly larger; thus,

$$y+\Delta y=(x+\Delta x)^2=x^2+2x\Delta x+(\Delta x)^2$$

If we subtract $y=x^2$ from the last expression we have

$\Delta y=2x\Delta x+(\Delta x)^2$, dividing by Δx, we have

$$\Delta y/\Delta x=2x+\Delta x.$$

In the limit as Δx approaches zero, the value of the above ratio, or the rate of change of our function, would be $2x$, as Δx would disappear. If our $x=100$, the above ratio would be 200, as determined above in the case of the numerical example. Another way of symbolizing the derivative is $D_x y=dy/dx$, but this requires a short explanation.

In Chapter XV we have already discussed the problem of the 'infinitesimal' and we have seen that 'infinitesimal' is a misnomer and that there is no such thing at all. Yet this word is very often uncritically used by mathematicians and is therefore often confusing. By an 'infinitesimal' mathematicians mean a *variable* which approaches zero as a limit. The condition that it should be a variable is essential. It would probably be better to call an 'infinitesimal' an *indefinitely* small quantity or 'indefinitesimal', and that is what the reader should understand when he sees anywhere the word 'infinitesimal' or 'infinitely small quantity'.

These indefinitely small quantities are in general neither equal, nor even of one order. Some by comparison are indefinitely smaller than others, and hence are said to be 'of higher order'. Usually several quantities are considered which approach zero simultaneously. In such a case one of them is chosen as the principal indefinitely small quantity. Let us recall that if we take any number, for example, 1, and divide it by 2 we have 1/2. If we divide 1 by 4 we have 1/4 which is smaller than 1/2; if we divide 1 by 10 we have 1/10 which is still smaller. If we carry this process on indefinitely, taking larger and larger denominators, the results are fractions of smaller and smaller values. In the limit, as the value of the denominator becomes indefinitely large the value of the fraction approaches zero. This simple consideration will help us in the classification of indefinitely small quantities.

Let us take a as the principal indefinitely small quantity and b another indefinitely small quantity. If the ratio b/a approaches zero with a we say that b is an indefinitely small quantity of higher order with respect to a. In other words, although a approaches zero in the limit yet it is infinitely larger than b and so the ratio b/a also approaches zero.

If the ratio b/a approaches a limit k different from zero as a approaches zero, then b is said to be of the 'same order' as a and $b/a=k+\epsilon$ where ϵ is indefinitely small with respect to a. In such a case $b=a\,(k+\epsilon)=ka+a\epsilon$, and ka is called the principal part of b. The term $a\epsilon$ is obviously of a higher order than a.

We may say in general that if we have a power of a, for instance a^n, such that the ratio b/a^n approaches a limit different from zero, b is called an 'infinitesimal' (indefinitesimal) of order n with respect to a.

Let us give a numerical illustration. We know that there are 60 minutes in one hour, 24 hours in a day, or that there are 1440 minutes in a day, and by multiplying 1440 by 7, that there are 10,080 minutes in a week. Our forefathers called this 1/10,080 part of a week a 'minute' because of its minuteness. It is obvious that a minute is very small as compared with a week. But if we subdivide a minute into 60 equal parts we have a still smaller quantity, a quantity of second order smallness and so we called it a second. Indeed there are 3600 seconds in one hour, 86,400 seconds in a day and 604,800 seconds in a week. If we decide that for some purpose a minute is as short a period of 'time' as we need to consider, then the second, 1/60 of a minute, is relatively so small that it could be neglected. In a calculation where 1/100 of some unit is the smallest value which needs to be considered, we may define this 1/100 as of first order

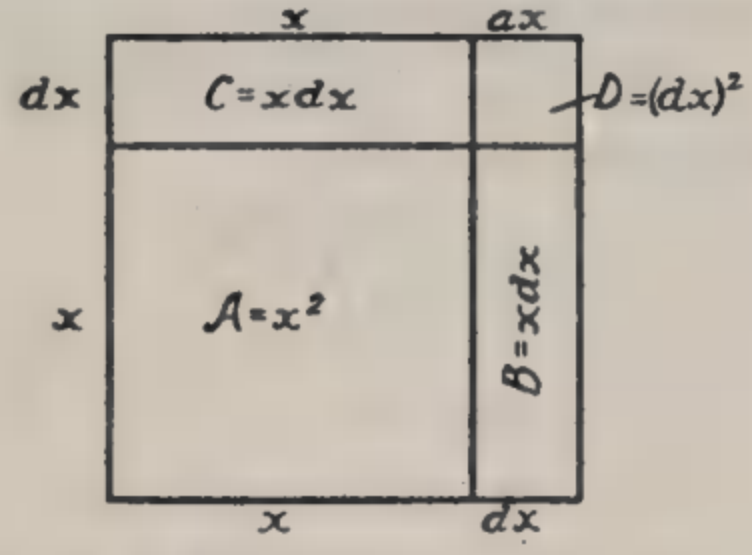


FIG. 1-A

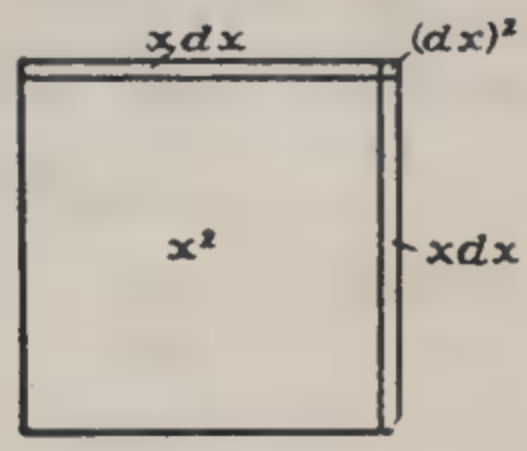


FIG. 1-B

smallness. Then 1/100 of 1/100, or 1/10,000, of that unit, which is relatively of second order smallness, is entirely negligible. The fractions whose smallness we are considering here are comparatively large, and we usually deal with much smaller quantities, but the smaller a quantity is, the more negligible the correspondingly smaller quantity of higher order becomes.

Let us consider a geometrical interpretation of the above. If we represent a quantity x by a line segment, and a slightly greater quantity, $x+dx$, by a slightly longer line segment; then the quantities x^2 and $(x+dx)^2=x^2+2xdx+(dx)^2$ may be represented by squares where sides are the line segments which represent the quantities x and $x+dx$ respectively.

If we denote the areas by A, B, C, D, we see that $A=x^2$ and that $A+B+C+D=x^2+2xdx+(dx)^2$. If we select our dx smaller and smaller the areas $B=C=xdx$ diminishing in one dimension only, become also smaller and smaller, but $D=(dx)^2$ is vanishing much more rapidly as it is diminishing in each of two dimensions, whence it is said to be a quantity of second order smallness, which for all purposes at hand may be neglected.

If we take $y=f(x)$ and its derivative

$$\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = D_x y.$$

Then $\dfrac{\Delta y}{\Delta x} = D_x y + \epsilon$, where ϵ is an indefinitesimal,

and $$\Delta y = D_x y \Delta x + \epsilon \Delta x.$$

In the above expression $D_x y\Delta x$ represents the principal part and $\epsilon\Delta x$ appears as an indefinitesimal of higher order. This principal part is called the differential of y and is denoted by dy. If we choose $f(x)=x$ we have $dx=\Delta x$ and so, $dy=D_x y\,dx$.

So we see that the differential of the independent variable x is equal to the increment of that variable. This statement is not generally true about the dependent variable, as ϵ does not generally vanish.

The derivative is also sometimes denoted as $f'(x)$ or y' and this notation is due to Lagrange; all three notations are used and it is well to be acquainted with them.

The derivative of a function $f(x)$ is in general another function of x, let us say $f'(x)$. If $f'(x)$ has a derivative, the new function is the derivative of the derivative or the *second derivative* of $f(x)$ and is denoted by y'' or $f''(x)$. Similarly the third derivative y''' or $f'''(x)$ is defined as the derivative of the second derivative and so on. In the other notations we have:

$$D_x(D_x y)=D_x{}^2 y \text{ or } \frac{d}{dx}\left(\frac{dy}{dx}\right)=\frac{d^2y}{dx^2}$$

Having introduced these few definitions it must be emphasized that the main importance of the calculus is in its central idea; namely, the study of a *continuous function* by following its history by *indefinitely small steps*, as the function *changes* when we give indefinitely small increments to the independent variable. As was emphasized before, the whole psycho-logics of this process is intimately connected with the activities of the *nervous structure* and also with the structure of science. In this work we are not interested in calculations, complications, or analytical niceties. Mathematicians have taken excellent care of all that. We need only to know about the structure and *method* which help to translate dynamic into static, and vice versa; to translate 'continuity' on one level, or order of abstraction, into 'steps' on another.

To illustrate what has been said and to give the reader the *feel* of the process, let us take for instance a simple equation $y=2x^3-x+5$ where y represents the function of the variable x expressed by a group of symbols to the right of the sign of equality.

To determine the relative rate of growth of this function, that is, to differentiate it, we replace x by a slightly larger value; namely, $x+\Delta x$, and see what happens to the expression. $2x^3$ becomes $2(x+\Delta x)^3=2x^3+6x^2\Delta x+6x(\Delta x)^2+2(\Delta x)^3$; $-x$ becomes $-x-\Delta x$ and the constant 5 remains unchanged. In symbols, $y+\Delta y=2x^3+6x^2\Delta x+6x(\Delta x)^2+2(\Delta x)^3-x-\Delta x+5$, where Δy represents the increment of the function and Δx represents the increment of the independent variable.

Subtracting the original expression $y=2x^3-x+5$ we get the amount by which the function has been increased, namely:

$$\Delta y=6x^2\Delta x+6x(\Delta x)^2+2(\Delta x)^3-\Delta x.$$

To determine the *relation*, or *ratio*, of Δy, the increment of the function, to Δx the increment of the independent variable which produced Δy, we divide Δy by Δx, and obtain the equation

$$\frac{\Delta y}{\Delta x} = 6x^2 + 6x\Delta x + 2(\Delta x)^2 - 1.$$

Then as Δx approaches 0 the terms in the right-hand side of the equation which contain Δx as a factor also approach 0 and replacing the left-hand side by $\frac{dy}{dx}$ we obtain the equation $\frac{dy}{dx} = 6x^2 - 1$ which means, that as Δx approaches 0, the ratio of the increment of the function to the increment of the independent variable approaches $6x^2 - 1$, true for any value we may arbitrarily assign to x.

It should be noticed that in our function the left-hand side represents the 'whole' as composed of interrelated elements which are represented by the right-hand side. When instead of x we selected a slightly larger value; namely, $x + \Delta x$, we performed upon this altered value *all* the operations indicated by our expression. We thus have in mathematics, because of the self-imposed limitations, the first and only example of *complete* analysis, impossible in physical problems as in these there are always characteristics left out.

An important structural and methodological issue should also be emphasized. In the calculus we introduce a 'small increment' of the variable; we performed upon it certain indicated operations, and in the final results this arbitrary increment disappeared leaving important information as to the rate of change of our function. This device is structurally extremely useful and can be generalized and applied to language with similar results.

It has been noticed already that the calculus can be developed without any reference to graphs, co-ordinates or any appeal to geometrical notions; but as geometry is an all-important link between pure analysis and the outside world of physics, we find in geometry also the psycho-logical link between the higher and lower orders of abstraction. But the appeal to geometrical notions helps *intuition* and so is extremely useful. For this reason we will explain briefly a system of co-ordinates and show what geometrical significance the derivative has

We take in a plane two straight lines $X'X$ and $Y'Y$, intersecting at O at right angles, so that $X'OX$ is horizontal extending to the left and right of O, and YOY', is vertical, extending above and below O, as a frame of reference for the locations of point, lines, and other geometrical figures in the plane. We call this a two-dimensional rectangular system of co-ordinates. This method may be extended to three dimensions, and our points, lines, and other geometrical figures referred to a three-dimensional rectangular system of co-ordinates consisting of three mutually perpendicular and intersecting planes.

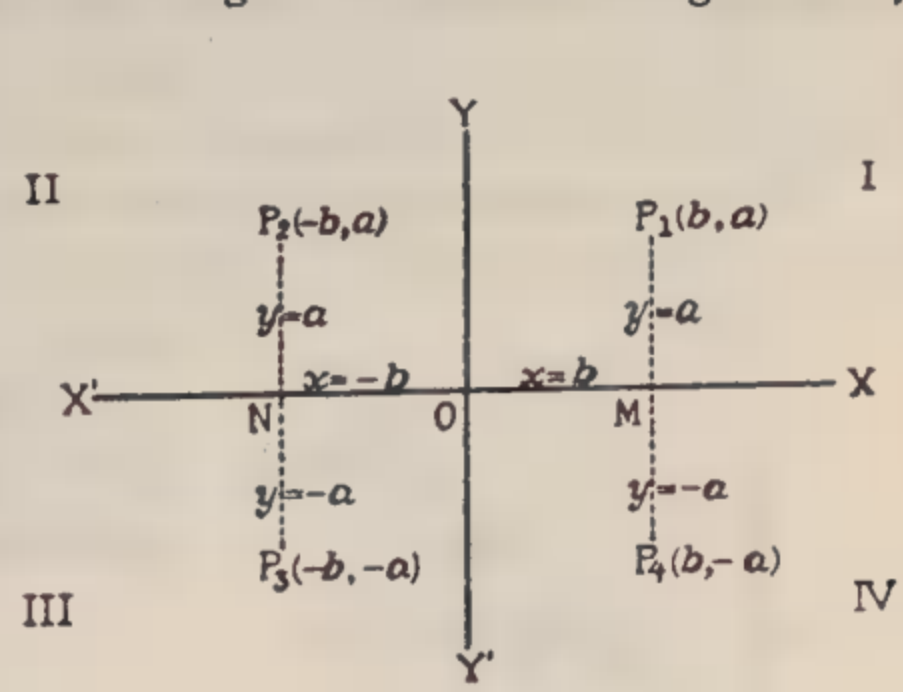


FIG. 2

As we see in Fig. 2, we have four quadrants I, II, III, IV, formed by the intersecting axes $X'X$ and $Y'Y$. The co-ordinates of a point P, by which we

mean the distances from the axes determine the position of the point uniquely. We call $X'X$ and $Y'Y$ the *axis* of X and the *axis* of Y respectively, and O the *origin*. If we select a point P_1 in the plane of $X'X$ and $Y'Y$ and draw a line P_1M perpendicular to $X'X$ then OM and MP_1 are called the co-ordinates of P_1; OM is called the *abscissa* and is denoted by $x=b$; and MP_1 is called the ordinate and denoted by $y=a$. We speak of P_1 as the point (b,a), or, in general, of any point as the point (x,y).

Let us draw $ON=OM=b$ and draw lines P_1P_4 and P_2P_3 through M and N respectively perpendicular to $X'X$, making $MP_4=NP_2=NP_3=MP_1=a$. We then have four points P_1, P_2, P_3, P_4, in each one of the four quadrants and all of them by construction would have equal numerical values for their abscissas and ordinates. To be able to discriminate between the four quadrants, and so avoid ambiguity, we make the convention that all values of y above $X'X$ are to be positive and below $X'X$ negative; and all values of x to the right of $Y'Y$ positive, to the left negative. Thus we see that by such conventions the point P_1 would have both b and a positive; P_2 would have b negative and a positive; P_3 both b and a negative, and finally P_4 would have b positive and a negative, or in symbols $P_1(b,a)$; $P_2(-b,a)$; $P_3(-b,-a)$; and finally $P_4(b,-a)$.

It is obvious that for any point on the X axis (for instance M) the ordinate $y=0$. If our point is on the Y axis the abscissa $x=0$ and the co-ordinates of the origin O are both zero (0,0).

From the above definitions we see at once how to plot, or locate, a point. To plot the point $(-4,3)$, since the abscissa x is negative and the ordinate y is positive we locate N on $X'X$, 4 units to the left of O. At N we erect a perpendicular upon which we locate the point $(-4,3)$, 3 units above N. The symbol $(-4,3)$ represents a particular case of the general symbol (x,y) and is accordingly plotted as a particular point as just shown. If instead of the pair or relations expressed by two equations $x=-4$, $y=3$, we have a single relation expressed by one equation, for example, $y=x-2$, we have y expressed as a function of x, whence by assigning to x different values, corresponding values of y are determined, and a set of points may be plotted where abscissas and ordinates are corresponding values of x and y respectively. Thus, when $x=0$, $y=-2$, when $x=1$, $y=-1$, when $x=2$, $y=0$, when $x=3$, $y=1$, when $x=4$, $y=2$, .

We may now plot the points $A(0,-2)$; $B(1,-1)$; $C(2,0)$; $D(3,1)$; $E(4,2)$; or as many more points as we may choose by giving x additional different values.

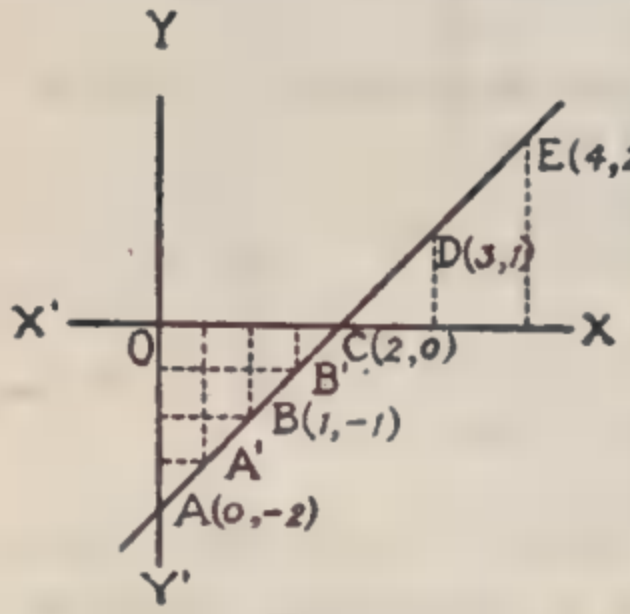


FIG. 3

If we give to x successive values with smaller differences our points would be closer together, for instance for

$x=0$	$y=-2$	(A)
$x=0.5$	$y=-1.5$	(A')
$x=1$	$y=-1$	(B)
$x=1.5$	$y=-0.5$	(B')
$x=2$	$y=0$	(C)
...	...	

As we plot larger and larger numbers of points closer and closer together, in the limit, if we take indefinitely many such points, we approach a smooth line. It can be proved that an equation of the type given in this example; namely, where both variables are of the first order, always represents a *straight* line. Such equations are called therefore *linear* equations, as they represent straight lines.

The problem of linearity and non-linearity is of extreme importance, and we will return to it later on. Here we are interested only in the definition and meaning of linearity of equations.

Let us consider next a simple equation of second degree, $y=\frac{x^2}{2}$. In assigning arbitrary values to x, we note that x^2 is always positive (by the rule of signs) whether x is positive or negative. Hence, we may tabulate values of x with the double sign $\pm$ meaning either $+$ or $-$.

$x=0$	$y=0$	(O)
$x=\pm 1$	$y=\frac{1}{2}$	(A)
$x=\pm 2$	$y=2$	(B)
$x=\pm 3$	$y=4\frac{1}{2}$	(C)
$x=\pm 4$	$y=8$	(D)
...	...	

We see for each value of y we have two values for x which differ only in sign This means that we have points on two sides of the Y axis with numerically equal abscissas and, since for $x=0$, $y=0$, the beginning of our curve is at the origin of co-ordinates and the curve is symmetrical with respect to the Y axis.

If we connect the points D', C', B', A', O, A, B, C, D, with straight lines we have a broken line. But if we choose smaller and smaller differences between the successive values of x, the broken line becomes smoother and smoother, and, in the limit, as we take increasingly smaller steps, or, in other words, plot indefinitely larger numbers of points in one interval, we approach a smooth, or continuous curve.

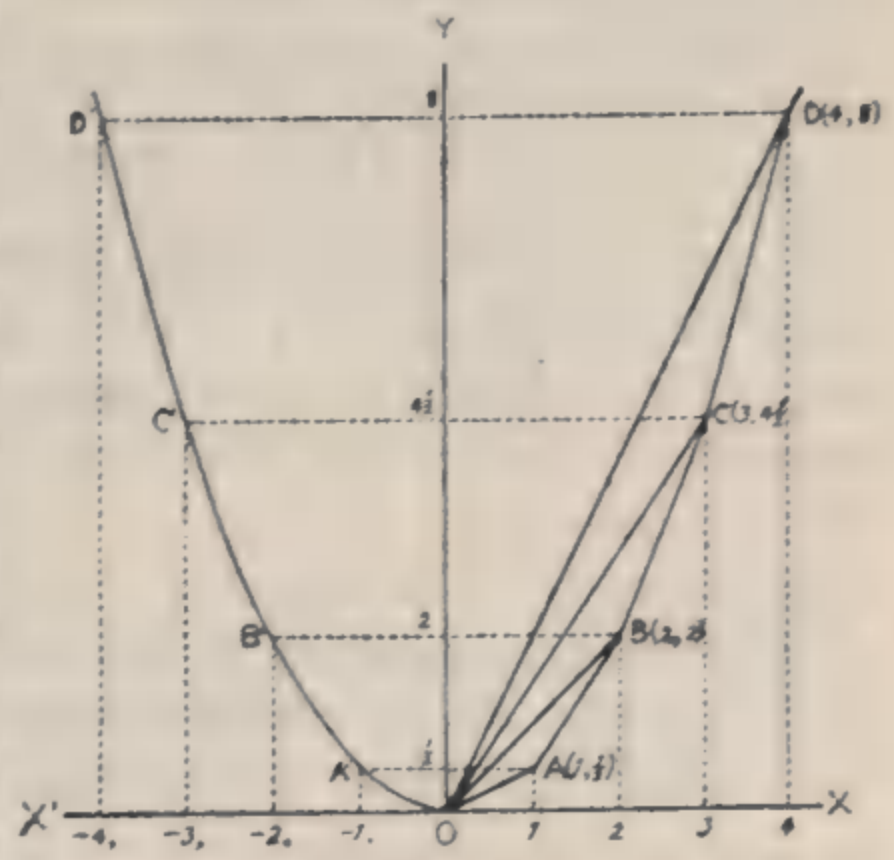


FIG. 4

It must be noticed that in equations of higher orders the ratio of changes in the function y to corresponding changes in the variable x vary from point to point, and so we have a *curve* instead of a straight line. It is necessary to become quite clear on this point so we may better compare the two different types of equations as to the law of their growth.

Let us write down in two columns the successive values for the two types of equations. Let us take the equation $y=\frac{x^2}{2}$ with the graph shown in the preceding diagram (Fig. 4) and the equation $y=2x$ as shown in Fig. 5.

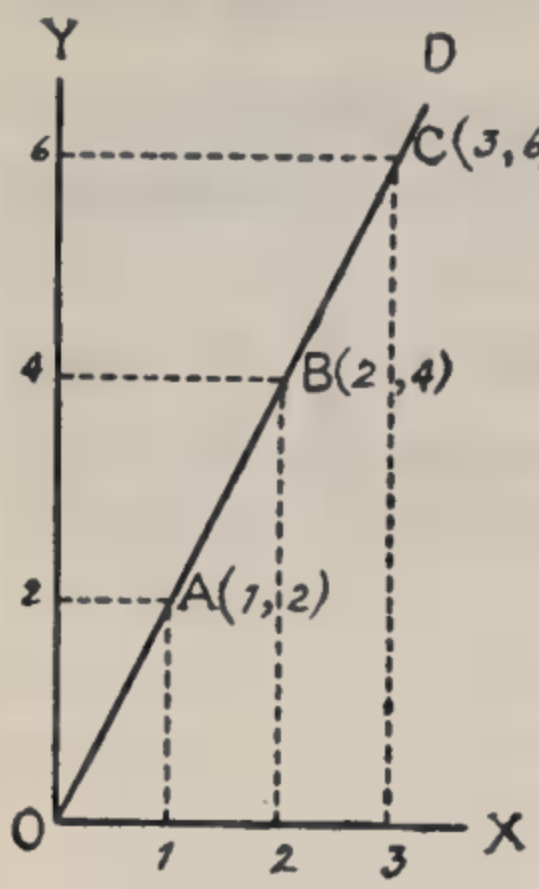


FIG. 5

Values of x	$y=2x$	$y=\frac{x^2}{2}$
−4	−8	8
−3	−6	$4\frac{1}{2}$
−2	−4	2
−1	−2	$\frac{1}{2}$
0	0	0
+1	+1	$\frac{1}{2}$
+2	+4	2
+3	+6	$4\frac{1}{2}$
+4	+8	8
+5	+10	$12\frac{1}{2}$
+6 , .	+12 , .	18 , .

The equation $y=2x$ involves the variables in the first degree and we see that the ratio of changes in the ordinates to corresponding changes in the abscissas remains constant (proportional). The triangles in Fig. 5, are either equal or similar, which necessitates the equality of angles and so the line $OABCD$ is of necessity a straight line. In this case as $x=0$ gave us $y=0$ the line passes through the origin of co-ordinates.

The picture is entirely different in the case of the higher degree equation, $y=\frac{x^2}{2}$, illustrated in Fig. 4. From the table of values of the function we see that the value of the function increases increasingly more rapidly than the values of the independent variable and so the ordinates are *not proportional* to the abscissas. If in Fig. 4 we connect O with A, O with B, O with C, O with D, respectively, we see that the lines OA, OB, OC, and OD have *different* angles with the axis $X'X$; the respective triangles are not similar, and so there is no proportionality. The lines OA, OB, OC, OD . , do *not* represent a straight line as they have all different angles with the axis XX' and so the points A, B, C, D . , cannot lie on a straight line but represent a *broken* line which, in the limit, when the points plotted become sufficiently near together, becomes a smooth and continuous curve.

The fact that equations in which the variables are only of the first degree, represent straight lines, and that equations of higher degrees represent curved lines is very important, as will appear later on. We must notice also that the problem of *linearity* is connected with *proportionality*.

These few simple notions concerning the use of co-ordinates will allow us to explain the geometrical meaning of the derivative and the differential.

Consider P_1 and P_2, (Fig. 6) two points on the curve, $y=f(x)$, referred to the axes OX and OY. Drop perpendiculars P_1M_1 and P_2M_2 from P_1 and P_2 to OX. These are the ordinates $y_1=f(x_1)$ and $y_2=f(x_2)$ of the points P_1 and P_2, and OM_1 and OM_2 are the abscissas x_1 and x_2 of the points P_1 and P_2. Through P_1 draw the secant P_1P_2, the tangent to the curve P_1T, and the line P_1Q parallel to OX. Then P_1Q represents $\Delta x=x_2-x_1$ the change in the variable x, and P_2Q represents $\Delta y=y_2-y_1=f(x_2)-f(x_1)$ the change in the function y.

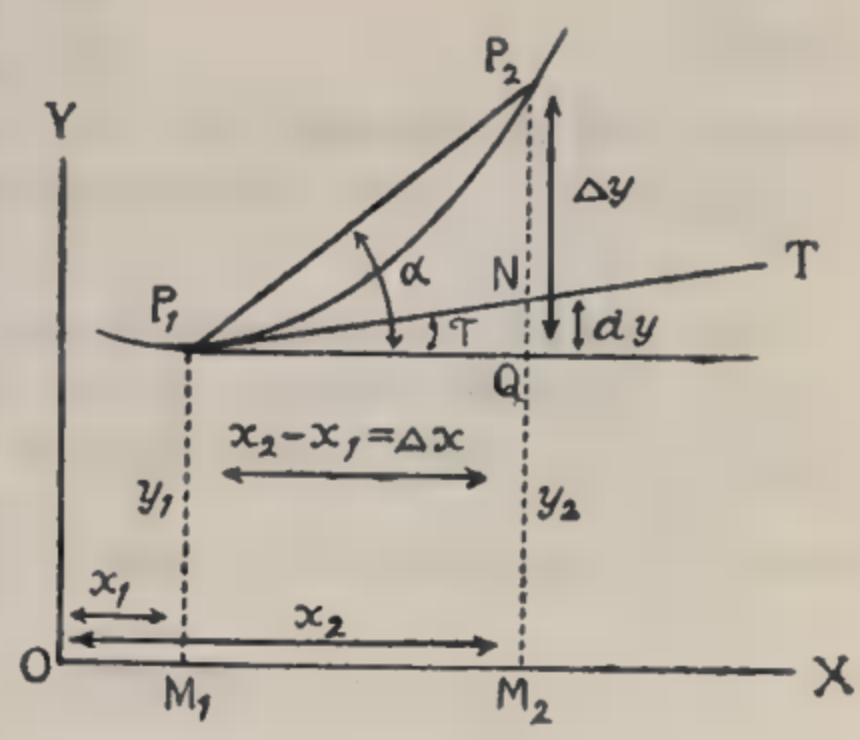


FIG. 6

In the right triangle P_1QP_2 the ratio P_2Q/P_1Q is a measure (the tangent) of the angle P_2P_1Q $(=\alpha)$ that is, $\tan\alpha$ $=P_2Q/P_1Q=\Delta y/\Delta x=\dfrac{f(x_2)-f(x_1)}{\Delta x}$ or,

since $x_2=x_1+\Delta x$

we may write $$\tan\alpha=\frac{f(x_1+\Delta x)-f(x_1)}{\Delta x}$$

As P_2 approaches P_1 along the curve, the secant P_1P_2 rotates about P_1 approaching P_1T as its limit, and the tangent of α approaches the tangent of τ, τ being the angle which P_1T, the tangent to the curve at P_1, makes with P_1Q. But as P_2 approaches P_1, $\Delta x=x_2-x_1=M_1M_2$ approaches zero or symbolically as $\Delta x\to 0$; $(\Delta y/\Delta x)\to\tan\tau$, that is $\tan\tau=\lim_{\Delta x\to 0}(\Delta y/\Delta x)$. We see that the $\lim_{x_2\to x_1}\dfrac{y_2-y_1}{x_2-x_1}=\lim_{\Delta x\to 0}\dfrac{\Delta y}{\Delta x}$ represents nothing more or less than the derivative of the function representing the curve. In other words, the geometrical interpretation of the analytical process of differentiation is the finding of the slope of the graph of the function. The increment Δy of the function is represented by P_2Q; the differential dy is equal to NQ and $\Delta x=dx=P_1Q$; $\tan\angle TP_1Q=\dfrac{dy}{dx}$.

From the above considerations we see that the differential calculus gives, by the application of some extremely simple structural principles, a method of analysis by which we can discover a tendency at a particular stage rather than the final outcome after a definite interval. From such fundamental yet simple beginnings the whole calculus is developed. Most of these developments are not needed for our purpose, but we will explain one specially important theorem. The theorem in question is that the derivative of the sum of two functions is equal to the sum of their derivatives. In symbols

$$D_x(u+v)=D_xu+D_xv.$$

Let us symbolize $u+v=y$ and select a special value

$$y_0=u_0+v_0 \tag{4}$$

then $y_0+\Delta y=u_0+\Delta u+v_0+\Delta v$. By subtracting (4), we have $\Delta y=\Delta u+\Delta v$. Dividing by Δx, we have $\frac{\Delta y}{\Delta x}=\frac{\Delta u}{\Delta x}+\frac{\Delta v}{\Delta x}$. When Δx approaches zero the left-hand side approaches $D_xy=D_x(u+v)$; and the first term of the right-hand approaches D_xu, while the second term approaches D_xv and so,

$$D_x(u+v)=D_xu+D_xv.$$

The symbol D_x means also that certain operations are to be performed upon our function; namely, to find its derivative. When used in this sense it is called an operator. The operator D_x can be also written in its differential form as $\frac{d}{dx}'$ and similarly for higher derivatives.

2. *MAXIMA AND MINIMA*

It will be useful to have some applications of the differential calculus explained.

If a function $y=f(x)$ is continuous in an interval $a<x<b$ and has larger (or smaller) values at some intermediate points than it has at or near the ends, then it has a maximum (or minimum) at some point $x=x_0$, inside this interval. If Fig. 7 represents the graph of the function, it is obvious that at the maximum (or minimum) the tangent to the curve is parallel to the axis and therefore the slope of this tangent is zero. As this slope is given by the derivative and the slope is zero we have a simple method of finding the maximum (or the minimum) of a function by equating the first derivative to zero; namely, $D_xy=0$ when $x=x_0$.

It is useful to be able to discriminate between the maximum and the minimum of a function. Fig. 7 shows that this can be done by finding means to discriminate between the two cases when our curve is concave upwards or concave downwards. The slope of a curve for a particular value of x is given by the value of D_xy, corresponding to that value of x. If the value D_xy is positive, y increases as x increases, and the curve slopes up as we move to the right; if the value of D_xy is negative, y decreases as x increases, and the curve slopes down as we move to the right.

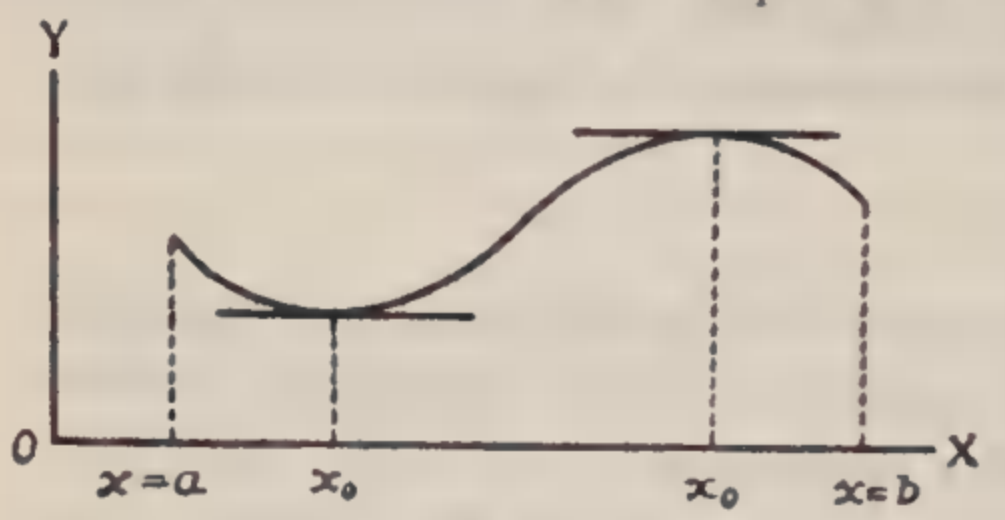


FIG. 7

If we consider the curve $y=f(x)$ which has its concave side turned upward (Fig. 8), the slope of the curve itself is a function of x, $\tan\alpha=f'(x)$. If we consider a variable point P on a curve $y=f(x)$, together with the tangent to the curve P, as following the curve in the direction of increasing values of x, the curve is concave upward whenever the slope is increasing algebraically,

that is when $D_x \tan \alpha = 0$. In other words, the curve is concave upwards for those values of x for which $D_x \tan \alpha$ is positive, or since $\tan \alpha = D_x y$ for those values of x for which $D_x \tan \alpha = D_x(D_x y) = D_x^2 y$ is positive. Similarly a curve is concave downwards for those values of x for which $D_x \tan \alpha = D_x(D_x y) = D_x^2 y$ is negative. These results can be expressed thus:

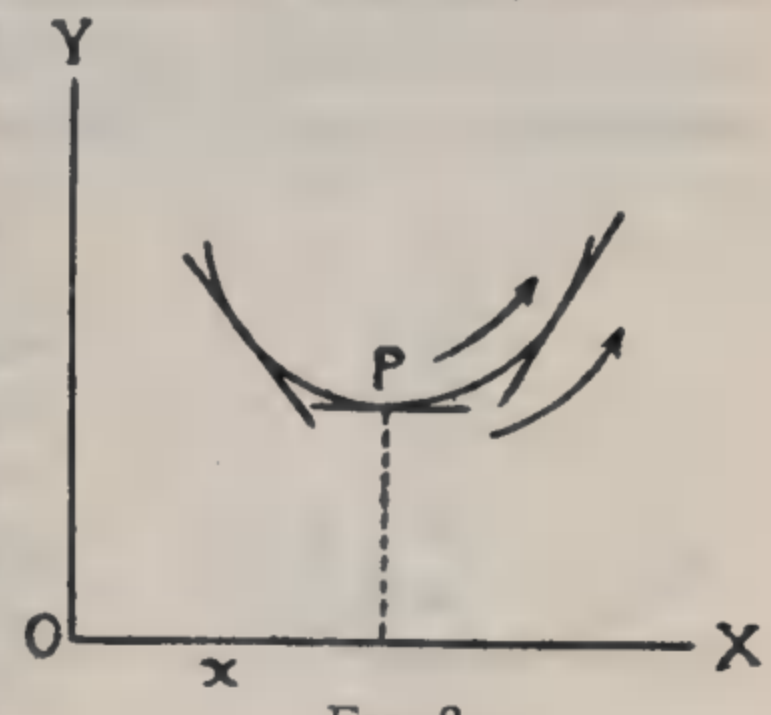


FIG. 8

A curve $y = f(x)$ is concave upward when $D_x^2 y > 0$, or, in words, when the second derivative is positive, and the curve is concave downward when the second derivative is negative, or, in symbols, when $D_x^2 y < 0$.

From Fig. 7, we see that for a maximum we must have our tangent parallel to the $X'X$ axis and our curve concave downwards, hence for these conditions the first derivative $[D_x y]_{x=x_0} = 0$, and the second derivative $[D_x^2 y]_{x=x_0} < 0$. For a minimum the first derivative must again be zero and the second derivative positive, whence the concave side of the curve is turned upwards. It should be noticed that the problems of maxima and minima play an extremely important structural psycho-logical and semantic role in our lives. All theories, somehow, are built on some minimum or maximum principle involving evaluations which are fundamental factors of all semantic reactions. In daily life we apply these structural and semantic notions continually. In science this tendency made its appearance quite early. The problem of maxima and minima was treated seriously as far back as the second century B.C. In the eighteenth century Maupertuis formulated a 'supreme law of nature', that in all natural processes the 'action' (energy multiplied by 'time') must be a minimum. Euler and Lagrange gave an exact basis and form to this principle; and finally Hamilton, in 1834, established this principle structurally as a *variational* principle, known as the hamiltonian principle, which appears to be of extreme generality and usefulness. It facilitates the derivation of the fundamental equations of mechanics, electrodynamics and electron theory. It has also survived, in a generalized form, the einsteinian revolution, for it contains nothing whatever which would connect it with a definite co-ordinate system; it involves only pure numbers and so is invariant to all transformations. It is structurally one of the most important invariants ascribed to nature, being independent of the systems of reference of the observers.

It is very desirable that this problem should be investigated further from the structural psycho-logical semantic and neurological point of view, as the very foundations of human psycho-logics are fundamentally connected with such a principle, which itself is an *invariant* in human psycho-logics.

Its importance is still increasing, and the hamiltonian principle plays a most remarkable role in all the newest advances of science. Any reader need only look attentively at his daily life to realize that there too this principle plays a predominant role.

3. *CURVATURE*

In modern scientific literature we hear often the fundamental term 'curvature' mentioned, and a few words about it will not be amiss. If we take two perpendicular lines $X'OX$ and OY and select on OY a number of points A, B, C, D . . , further and further away from O and describe arcs of circles with these points as centres with radii AO, BO, CO, DO . . , respectively Fig. 9, we find each successive arc flatter and closer to the line $X'X$ than its predecessor. In other words, the larger the radius of our circle, the flatter its arc is. In the limit as the radius of the circle becomes indefinitely large, the arc approaches a straight line by intuition and by definition. We notice also that the curvature of each circle is uniform, that is, one-valued at every point; but that when we pass from one circle to another of different radius, the curvature changes.

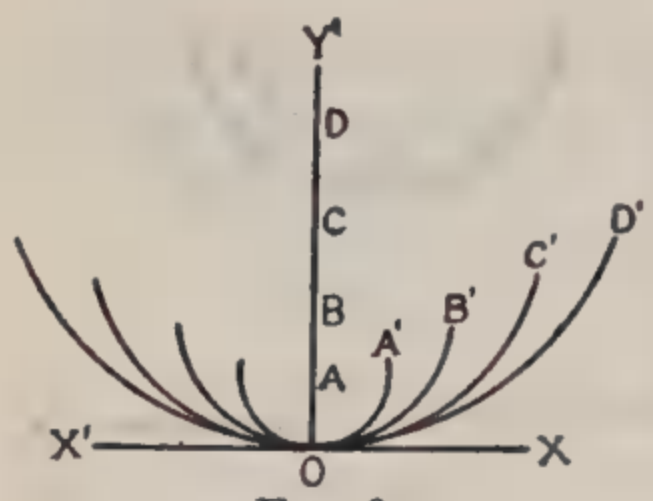


FIG. 9

If we consider a curve and two points on it, M_1 and M_2, (Fig. 10) and draw two tangents at these points; then the angle between these two tangents will depend on two factors, the sharpness of the curve and the distance between the points M_1 and M_2. If we take the points near enough and designate the length of the arc between them by Δs, the angle between the two tangents by $\Delta\theta$, then the limiting value of the ratio $\Delta\theta/\Delta s$, as M_2 approaches M_1, becomes $d\theta/ds$, and is a measure of the rate of change of the direction of the tangent at M, as M moves along the curve. Let us designate the rate at which the tangent turns where the point describes the curve with unit velocity as the curvature, or $k=\pm d\theta/ds$, but as k is essentially a positive number or zero we accept only the absolute value of this ratio. To find $d\theta/ds$ we notice that $\tan\theta=dy/dx$

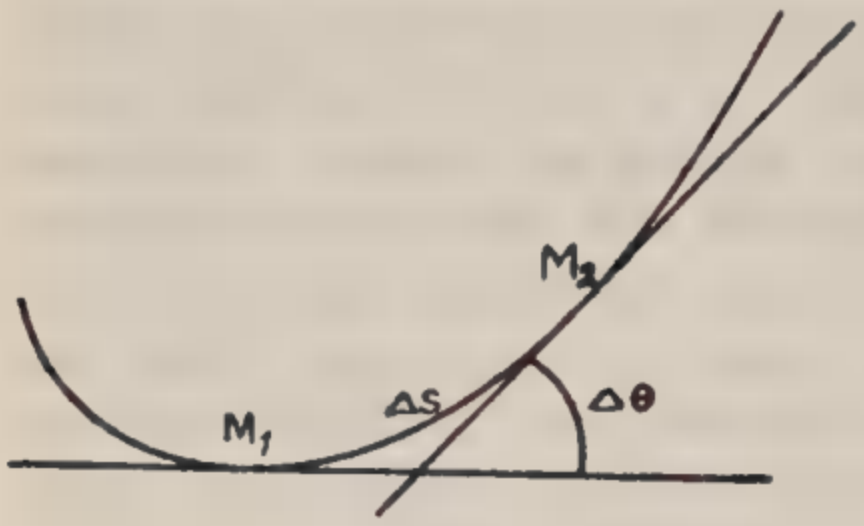


FIG. 10

or $\theta=\tan^{-1}\dfrac{dy}{dx}=\tan^{-1}y'$, whence $d\theta=\dfrac{dy'}{1+y'^2}=\dfrac{y''dx}{1+y'^2}$.

But $k=\dfrac{d\theta}{ds}$ where $ds=\sqrt{(dx)^2+(dy)^2}=\sqrt{1+y'^2}dx$

whence $k=\dfrac{y''}{(1+y'^2)^{3/2}}$.

The reciprocal of the curvature is called the radius of curvature. The radius of curvature of a circle is its radius. The curvature of a curve is measured by the radius of the osculating circle, that circle which fits the curve the most closely in the neighbourhood of our point.

4. *VELOCITY*

Until now we have treated our independent variables as *any* quantity: but there are many problems where the independent variable represents 'time'. For instance, if we travel by railroad the distance increases as 'time' increases, plants and animals grow with 'time', . By the average velocity with which a given point moves for a given length of 'time' we mean the distance s traversed divided by the 'time' elapsed. If, for instance, a train makes 5 miles in 10 minutes we say that its average velocity is 30 miles per hour, or, in symbols: Velocity, $v=\frac{s}{t}$. In this case we were considering uniform velocity, but very often we have to deal with velocities which are not uniform and which might be increasing or decreasing. In such a case we can describe the velocity approximately at any given moment if we take a short interval of 'time' immediately after the moment in question and take the average velocity for this short interval.

For instance, the distance a stone falls is according to the law, $s=16\,t^2$. We want to find how fast it is going after t_1 seconds when $s_1=16\,t_1^2$, and a short interval after we have, let us say, $s_2=16\,t_2^2$. Obviously the average velocity for the interval t_2-t_1 is $\frac{s_2-s_1}{t_2-t_1}$ feet per second. If we take $t_1=1$, $s_1=16$, and the difference $t_2-t_1=0.1$ of a second then $s_2=16\,t_2^2=16\times1.21=19.36$ and $\frac{s_2-s_1}{t_2-t_1}=\frac{19.36-16}{0.1}=\frac{3.36}{0.1}=33.6$ ft. per second.

If we take the interval of 'time' smaller, for instance, 1/100 of a second we would have $\frac{s_2-s_1}{t_2-t_1}=32.2$ feet per second, and if we take the intervals as 1/1000 of a second the average velocity would be 32.0 feet per second. We see that we could determine the speed of the stone at any instant with any degree of accuracy by direct calculation, but this is not necessary. If we regard the interval t_2-t_1 as an increment of the variable t, that is as Δt, and $s_2-s_1=\Delta s$ which represents the increment of the distance considered as a function of the 'time' we would have the average velocity $=\Delta s/\Delta t$. As Δt approaches zero in the limit, the average velocity approaches a limit and this limit is the velocity v at the instant t_1, or in symbols

$$v=\lim_{\Delta t\to 0}\frac{\Delta s}{\Delta t}=\frac{ds}{dt}.$$ In words, the velocity of a point is the 'time' derivative of the space traveled.

If the velocity is not uniform, the rate at which the velocity is increasing is called the acceleration and may be written as $a=\frac{dv}{dt}$, but as we have already seen dv is itself $d\left(\frac{ds}{dt}\right)$, hence $a=\frac{d^2s}{dt^2}$. In words, the acceleration is the second derivative of the distance with respect to 'time'.

In the above notes we have not attempted to give the reader more than some structural and methodological notions, and what amount really to short structural explanations of definitions which will be useful later on. The reader can find many excellent books which give all the additional information he may want.

Section C. On the integral calculus.

So far, we have been studying a method by which to find the variation of a given function corresponding to an indefinitely small variation of our variable. We saw that the *rate of change* of our function was given by the first derivative, which in turn was also a function (usually different) of our independent variable and so could itself vary and have a rate of change, and so give us a second derivative , .

And now we must explain briefly the inverse problem; namely, given the derivative to find the function. In symbols, given $u = D_x U$, find U.

The function U is called the integral of u with respect to x, or, in symbols, $U = \int u dx$.

To integrate a function $f(x)$ is to find a function $F(x)$ which when differentiated gives again the function $f(x)$ with which we started. As in this work we are not interested in computations, but only in the structural, methodological, and semantic aspects, the inverse problem of differentiation; namely, integration, is less important for us here, and I will explain only a single example. We have already differentiated the function $y = 2x^3 - x + 5$ and found its derivative $dy/dx = 6x^2 - 1$. Just as the derivative of the sum of a number of functions is equal to the sum of their derivatives, a similar rule holds for the integrals; namely, that the integral of the sum of a number of functions is equal to the sum of their integrals. Hence we can take in our example only the first term of our equation. In symbols $D_x(2x^3) = 6x^2$; in words, the derivative of $2x^3$ is $6x^2$.

In a problem in integration we would have $6x^2$ given and we would have to find the original function from which $6x^2$ was obtained by differentiation. In our case the solution is already given; namely, $\int 6x^2 dx = 2x^3$.* In general the solution of problems of integration is largely dependent on the ingenuity of the solver, although we have a number of standard formulae and methods. The geometrical meaning of integration is much more interesting for us and we will give a short explanation of it.

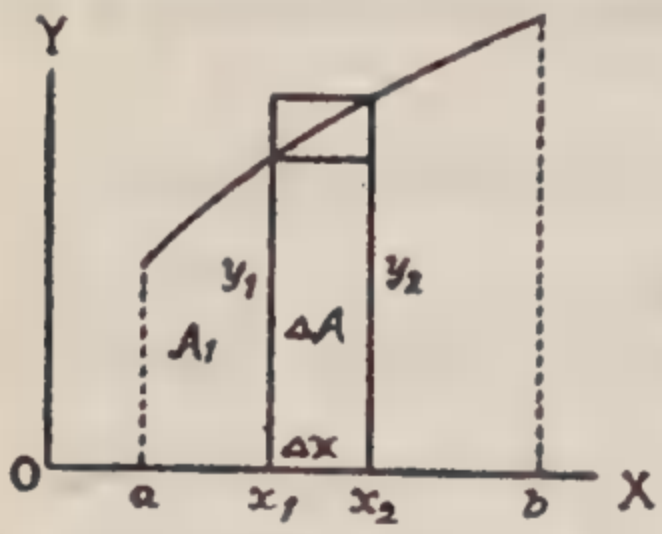


FIG. 11

If we consider the curve given by an equation $y = f(x)$ and the area bounded by the X axis, the two ordinates whose abscissas are $x = a$ and $x = b$ and the curve, we may find the area as follows:

*The constant of integration is omitted so as not to confuse the reader.

If we select an arbitrary value $x=x_1$ for which $y=y_1=f(x_1)$, denoting the corresponding value of the area A by A_1 (Fig. 11) and give to x_1 an increment Δx, then the area A_1 would receive the increment ΔA. We can approximate ΔA by the help of two rectangles, one of height $y_1=f(x_1)$, the other of height $y_2=y_1+\Delta y=f(x_2)=f(x_1+\Delta x)$.

We see that ΔA is larger than the smaller rectangle. In symbols

$$y_1\Delta x<\Delta A<(y_1+\Delta y)\Delta x,$$

hence

$$y_1<\frac{\Delta A}{\Delta x}<(y_1+\Delta y).$$

As we pass to the limit and let Δx approach zero, we have $\lim_{\Delta x\to 0}\frac{\Delta A}{\Delta x}=y_1$. That is, $D_xA=y_1=f(x_1)$ when $x=x_1$; which means that the ordinate of the curve at any point is equal to the x derivative of the area at that point. In general, $D_xA=y$, and hence, $A=\int ydx$.

The consideration of what is called the definite integral is still more instructive. Let us take the curve in Fig. 12 represented by an equation $y=f(x)$ and a pair of ordinates which intersect the X axis at the points $x=x_0$ and $x=x_n$. Let us divide the interval x_0x_n into n equal parts and erect ordinates at each point of division. Let us construct a set of pairs of rectangles with these ordinates as we constructed the single pair of rectangles in Fig. 11. By inspection of the figure we see that the area under the curve is slightly greater than the sum of the areas of the included rectangles and slightly less than the sum of the areas of the including rectangles. When n is allowed to increase without limit the sum of the areas of either set of these rectangles approaches the area bounded by the curve, the X axis, and the end ordinates. In symbols, the area of the first rectangle beneath the curve is $f(x_0)\Delta x$, where Δx denotes $x_1-x_0=\frac{x_n-x_0}{n}$. The area of the second rectangle is $f(x_1)\Delta x$, . The sum of these areas is $f(x_0)\Delta x+f(x_1)\Delta x+\ldots+f(x_{n-1})\Delta x=\sum_{i=0}^{n-1}f(x_i)\Delta x$.

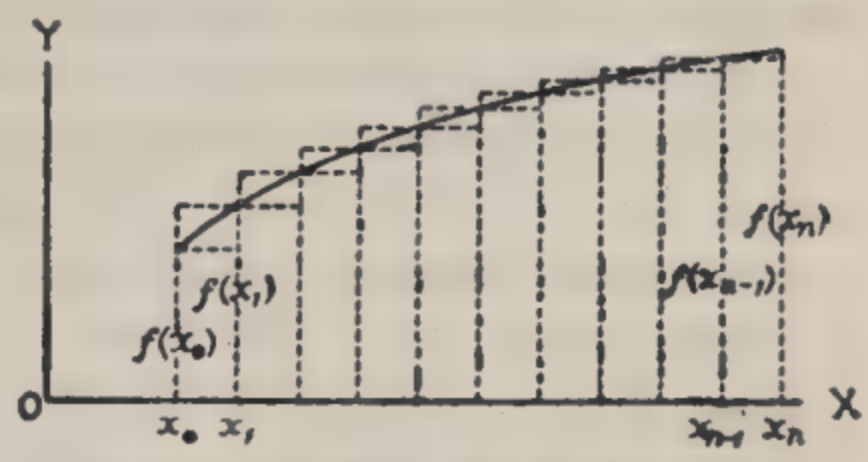


FIG. 12

If we allow n to increase without limit we have the area under the curve:

$$A=\lim_{n\to\infty}[f(x_0)\Delta x+f(x_1)\Delta x+\ldots+f(x_{n-1})\Delta x]=\lim_{\Delta x\to 0}\sum f(x)\Delta x$$

$$=\int_{x=x_0}^{x=x_n}f(x)dx=\Big[F(x)\Big]_{x=x_0}^{x=x_n}=F(x_n)-F(x_0).$$

In words, the above formula indicates the fundamental process of the integral calculus; namely: Let $f(x)$ be a continuous function of x throughout the interval $x_0\leqq x\leqq x_n$. If we divide this interval into n equal parts by the points $x=x_0, x_1, \ldots, x_n$, and form the sum $f(x_0)\Delta x+f(x_1)\Delta x+\ldots+f(x_{n-1})\Delta x$,

as we let n increase without limit, this sum will approach a limit, which can be found by integrating the function $f(x)$, that is, by finding the function $F(x)$ of which $f(x)$ is the derivative, and by taking the integral between the limits $x=x_0$ and $x=x_n$; that is, by taking the difference between $F(x_n)$ and $F(x_0)$.

It must be noticed that in our first example, the case of the indefinite integral, we considered integration as the inverse of differentiation; in the second example, we considered the definite integral as the limit of a sum.

The symbol of the integral, $\int$, had its origin in the letter S from the latin word 'summa', the integral being historically understood as the definite integral, or the limit of a sum.

Section D. Further applications.

1. *PARTIAL DIFFERENTIATION*

When we have more than one independent variable, for example, two, we have to become acquainted with what is called partial differentiation. This process is important, as in practice we usually deal with several independent variables. It presents very little that is new from a structural and methodological point of view, but we give it here, simply to explain the meaning of the term, as the reader may find it used in other works.

If we have a function z of two independent variables x and y, $z=f(x,y)$ which geometrically represents a surface, we may differentiate with respect to one of the variables, let us say x, and hold the other variable y fast, that is, treat it as a constant. In this way we should then have a partial derivative of z with respect to x. Similarly, if we treat x as a constant and differentiate in respect to y, we should have the partial derivative of z with respect to y. The above definitions give us the rules for partial differentiation—that is, following the ordinary rules, considering each variable individually and treating all the other variables as constant.

The notation for partial derivatives is similar to the ones explained before, except that the lower case letter d is replaced by the script form ∂ or a subscript is used to indicate the variable with respect to which the differentiation is performed; for instance, $\frac{\partial f}{\partial x}=\frac{\partial z}{\partial x}=f_x'=z_x'=D_xf=D_xz$, . Higher derivatives are obtained without difficulty in like manner. If $z=f(x,y)$ and $\frac{\partial z}{\partial x}=f_x'(x,y)$ and $\frac{\partial z}{\partial y}=f_y'(x,y)$, the partial derivatives themselves are in general also functions of x and y and can in turn be differentiated. Thus, $\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right)=\frac{\partial^2 z}{\partial x^2}=f_{xx}''(x,y)$, or $\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)=\frac{\partial^2 z}{\partial x\partial y}=f_{xy}''(x,y)$. The order in which we differentiate is immaterial provided that the derivatives concerned are continuous. The *total differential* of a function of two variables, for example, $f(x,y)$, $df=d_xf+d_yf=\frac{\partial f}{\partial x}dx+\frac{\partial f}{\partial y}dy$ is

equal to the sum of the partial differentials of the first order if we neglect terms of higher orders, whose values are indefinitely small quantities relative to the first order differentials. In symbols, $df = d_x f + d_y f = \left(\frac{\partial f}{\partial x}\right)dx + \left(\frac{\partial f}{\partial y}\right)dy$. In words, the total differential of $f(x,y)$ is found by finding the partial derivatives with respect to x and y, multiplying them respectively by dx and dy, and adding.

2. *DIFFERENTIAL EQUATIONS*

A natural development of the invention of the calculus was the introduction of differential equations. Differential equations differ from the ordinary equations of mathematics in that in addition to variables and constants they contain also derivatives of one or more of the variables involved. Differential equations are of extreme importance, and arise in many problems. Newton solved his first differential equation in 1676 by the use of an infinite series, eleven years after his discovery of the calculus in 1665. Leibnitz solved his first differential equation in 1693, the year in which Newton first published his results. From this date on, progress in the development and application of differential equations was very rapid, and today the subject of differential equations occupies in the general field of mathematics a central position from which important and useful lines of development flow in many different directions.

To integrate or solve a differential equation means, analytically, to find all the functions which satisfy the equation. In geometry, it means to find all the curves which have the property expressed by the equation. In mechanics it means to find all the motions that may possibly result from a given set of forces , . The *degree* of the differential equation is defined as the degree of the derivative of the highest order which enters the equation. The *order* of the differential equations is the order of the highest derivative it contains.

Equations in x and y, of the first degree in y and its derivatives with respect to x, y', y''. , are called *linear equations*. The main equations of physics are *linear* differential equations of the second order, since y, the primitive function, y', the first derivative, and y'', the second derivative, appear only in the first degree. For instance the equation $\frac{d^2y}{dx^2} + a_1\frac{dy}{dx} + a_2y = X$, or $y'' + a_1y' + a_2y = X$, when X represents a function of x alone is such an equation. It is linear, or of the first degree, because the second derivative, y'', appears only to the first degree. It is of the second order because that is the highest order derivative in the equation. As we may recall, the derivative of a function gives us the *rate of change* of the function when we give successive values to the independent variable. When we study the rate of change of the rate of change of our function, we study the rate of change of the first derivative which expresses the rate of change of the function, whence we obtain the derivative of the second order, and so on. If we equate our derivatives to zero, or choose a value of the variable for which our derivative becomes zero, the rate of change of our function

becomes zero. In other words, the value of our function is momentarily constant, it has a stationary value.

Quite naturally, differential equations which involve derivatives involve implicitly and explicitly the whole fundamental structural framework of the calculus as explained in this chapter by expressing the 'rate of change' of some natural process. If the rate of change is zero, it might express some 'natural law', or some uniformity as found in nature. In other words, differential equations express differential laws, which in turn express the momentary tendencies of processes whose outcomes are given by the process of integration.

From what has already been said here, it is obvious that differential equations and the differential laws which they express are of extreme structural importance. They formulate not only the uniformities and tendencies found in nature, but also of necessity somehow involve causality. Besides which, they are also in accord with the physical structure and function of the nervous system. We shall return to this most important subject in the next chapter, in which we shall analyse the physical significance and aspects of what has been explained here.

3. *METHODS OF APPROXIMATION*

In discussing the above fundamental notions of the calculus we considered a portion AB, of the curve given by the equation $y=f(x)$, (Fig. 13) and two points on this curve P_1 with co-ordinates (x_1, y_1) and P_2 with co-ordinates (x_2, y_2) moving along the curve, the secant, or chord, P_1P_2 rotates about P_1, its length steadily diminishing, and in the limit as the length of the chord P_1P_2 tends toward zero, the slope of the secant approaches the slope of the tangent P_1T. We saw that the slope of this tangent was given by the value of the first derivative of the function which represented the curve. We were trying to get some knowledge of the direction of our curve at a given point by considering the slope of a *straight line* of smaller and smaller length. When we studied the curvature of our curve we considered the rate of change of the slope of our tangent and so, by the aid of a second derivative, we found the curvature. In this case we approximated our curve to a circle of radius equal to the radius of curvature of the curve at a given point.

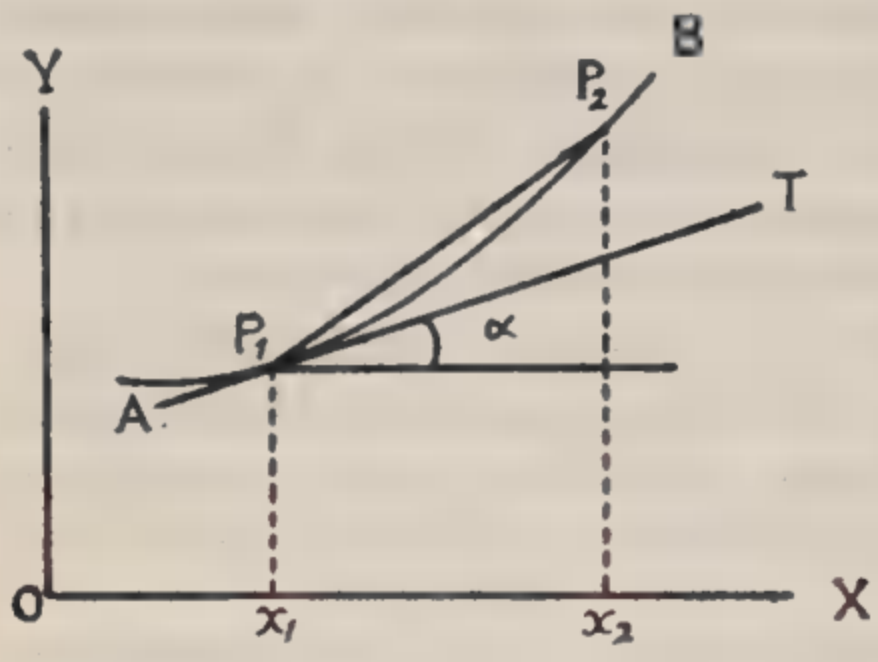


FIG. 13

In attempting to determine the length of a portion of our curve a point cannot be regarded as a piece of the curve but only as marking a position on it. For the purpose of determining the length of an arc it is convenient to replace each small element of the arc by its chord, a *lineal element*. By definition the length of an arc of a curve is the limit, if such limit exists, toward which the

sum of the lengths of the chords of its small subdivisions tends as the number of chords increases indefinitely and their individual lengths all approach zero uniformly. For example, the circumference of a circle is the limit approached by the perimeter of an inscribed polygon as the number of its sides increases indefinitely, the lengths of the individual sides all approaching zero.

Similarly the length of a curve may be approximated by the sum of the lengths of segments of tangents at successive arbitrarily chosen points, merely by choosing the points nearer and nearer together. For example, the circumference of a circle is the limit approached by the perimeter of a circumscribed polygon as the number of its sides increases indefinitely, the lengths of the individual sides all approaching zero. In either case, a point on a curve taken with a vanishingly small portion of the tangent to the curve at that point may be called the *lineal element* of the curve.

The above definitions apply equally well in either two or three dimensions. The lineal element in two dimensions may be defined by three co-ordinates x, y, p, of which x and y are the co-ordinates of the point through which the lineal element passes and p is the slope of the element. This slope, as we already know, is to be found by differentiation, and is given by the formula $p = dy/dx$. In geometrical problems which relate the slope of a tangent to that of other lines, it is not the tangent that is of real importance but the *lineal element.* From this point of view a curve is made up of infinite numbers of vanishingly small lineal elements which are tangent to it, which is the point of view of the differential calculus. Or the curve is composed of infinite numbers of vanishingly small chords which are the sides of an inscribed polygon, which is the point of view of the integral calculus.

Obviously, in the limit, both points of view are equivalent, although as a matter of convenience they may be different. In any case, it must be obvious to the reader that using *straight lines* instead of pieces of a curve, or using as closer approximations arcs of circles, facilitates our study of the curves, indeed renders such study possible at all, and in practice we can carry our work to any degree of approximation we choose. But in theoretical work we require precision, hence we think in terms of infinite numbers of vanishingly small steps. The differential and integral calculus supply the only perfect technique for these processes of analysis and synthesis.

4. *PERIODIC FUNCTIONS AND WAVES*

We have already said that the most important relations of physics are represented by linear differential equations of the second order. It is important to know the connection of these equations with the general theory of waves or oscillations.

If on a circle of unit radius, as shown in Fig. 14, we take several points P_1, P_2, P_3, P_4, and connect these points by straight lines with the centre O, we get angles XOP_1, XOP_2 , . In trigonometry we define certain functions of these angles and a unit of measurement. For our purpose we will only define the so-called sine and cosine, as we have already met the definition of tangent

$\tan\theta = \frac{M_1P_1}{OM_1}$. The angles XOP_1, XOP_2 . , may be specified by the ratios M_1P_1/OP_1, M_2P_2/OP_2 . , respectively each of which ratios has a definite value. This ratio in any case is called the *sine* of the angle, and is written in abbreviated form $\sin\theta$. If the radius of our circle is taken as unity then simply $M_1P_1 = \sin \angle P_1OM_1 = \sin\theta$, since $OP_1 = 1$. The ratio OM_1/OP_1 is called the *cosine* of the angle XOP_1, and is written $\cos\theta$.

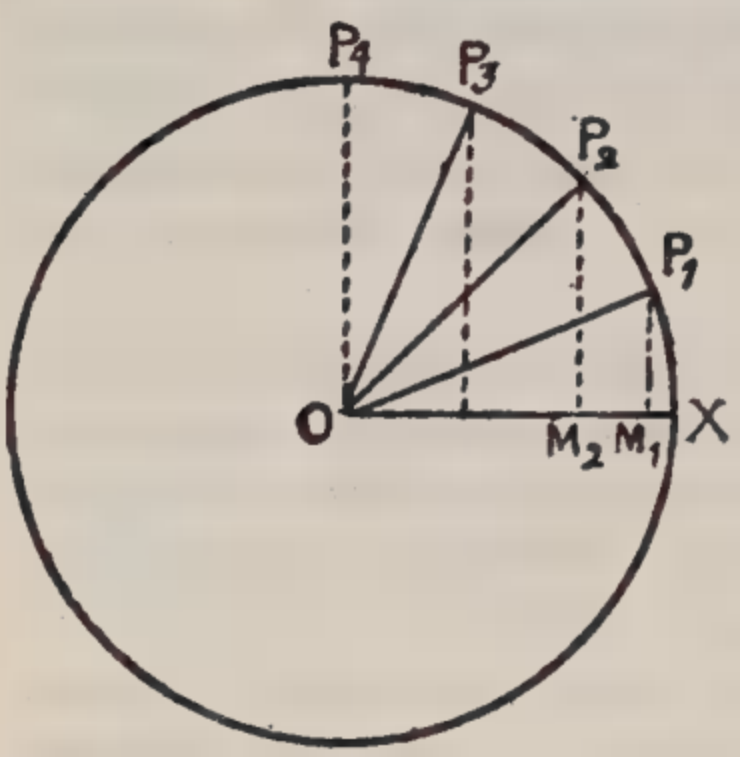


FIG. 14

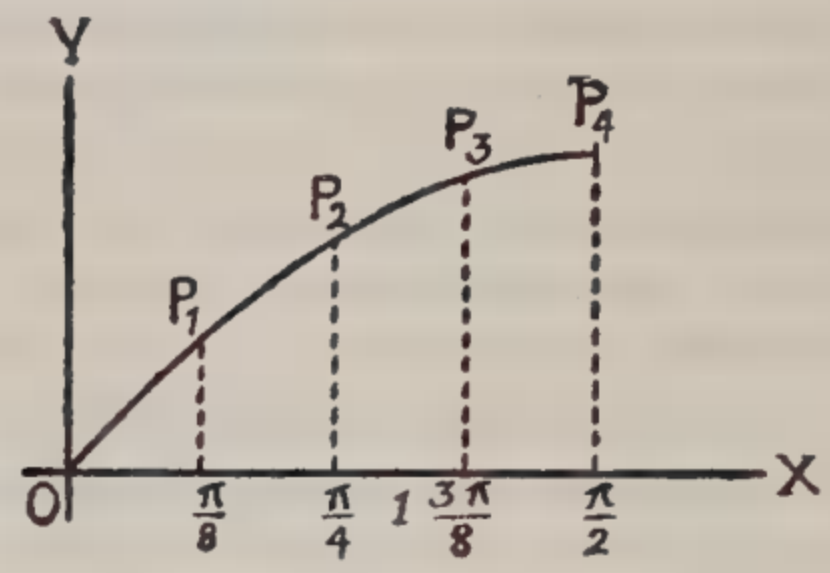


FIG. 15

There are two units of measurement of angles. In ordinary, or sexagesimal, measure, the unit angle is the degree, 1/360 of the entire angle about a point, 1/180 of a straight angle, or 1/90 of a right angle. The degree is divided into 60 equal parts called minutes. The minute is divided into 60 parts called seconds. In circular measure the unit angle is the radian, the angle at the centre of a circle whose arc is equal to the radius of the circle. This angle is a constant whether the circle be large or small, due to the fact that the circumferences of circles vary as their radii, and, in one circle, angles at the centre are proportional to their arcs. The constant ratio of the circumference of the circle to its radius is given by the number $\pi = 3.14159\ldots$, this number being 'incommensurable' with unity. As the length of the circumference of a circle with radius R, is $2\pi R$ we see that the entire angle about the centre, which in degrees is 360, is in radians 2π; that a straight angle equals 180 degrees or π radians; and that a right angle equals 90 degrees or $\frac{\pi}{2}$ radians.

Thus 1 radian $= \frac{180^\circ}{\pi} = 57^\circ 17' 44''.806\ldots$ which, as it depends on the value of π, is itself an 'irrational' number. The 'incommensurability' of the radian with right and straight angles makes its practical use inconvenient. One of the main uses of the radian is in theory as it introduces a marked simplification in that the ratio of the sine of an indefinitely small angle to the angle itself is 1, when the angle is measured in radians. In other words, the equivalence of an indefinitely small arc and chord becomes apparent numerically when the angle and sine are expressed in one unit.

The following table gives the ordinary and radian measures, the sine, cosine and tangent of angles of 0, 1, 2, 3, and 4 right angles.

Angle in Right Angles	Angle in Degrees	Angle in Radians	Sine	Cosine	Tangent
0	0	0	0	1	0
1	90	$\pi/2$	1	0	$\pm\infty$
2	180	π	0	-1	0
3	270	$3\pi/2$	-1	0	$\mp\infty$
4	360	2π	0	1	0

From Fig. 14 and from this table, it follows that the values of the trigonometric functions are equal for the angles 0 and 2π, or in the language of degrees, for the angles 0° and 360°. We see also from Fig. 14 that the angle XOP_1, or any other angle, has one measure as expressed by its trigonometric functions if we add to it 360° or 2π radians.

The structural importance of the trigonometric functions in analysis lies in the fact that they are the *simplest* singly *periodic* functions and are therefore adapted for the representation of undulations. As we have already seen the sine and cosine have the single real period 2π, which means that they are not altered in value by the addition of 2π to the variable. The tangent has the period π.

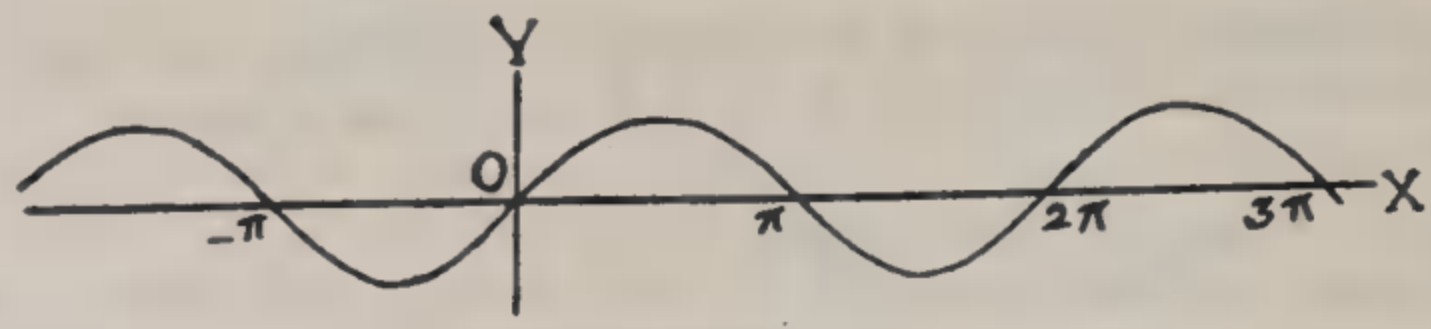
Besides the three functions defined above, we usually define three others, the secant, the cosecant and the cotangent as reciprocals respectively of the cosine, the sine, and the tangent. These last three we may disregard in our present discussion.

Let us consider the function $y = \sin x$, and construct the curve which this equation represents. If we draw a circle of *unit* radius, Fig. 14, the ordinates corresponding to the different angles XOP_1, XOP_2 . , give the values of y, while the angles measured in radians, give the corresponding values of the abscissa x.

Plotting corresponding values of x and y as thus obtained in Fig. 14 we get in Fig. 15 the partial graph of the function $y = \sin x$. Proceeding again around our circle in Fig. 14, that is, adding 360° or 2π, to each of our angles, hence to their abscissas of the curve in Fig. 15, we add to the graph a second complete wave. We may thus proceed either forward or backward obtaining as many complete waves, or undulations, as we please, as in Fig. 16.

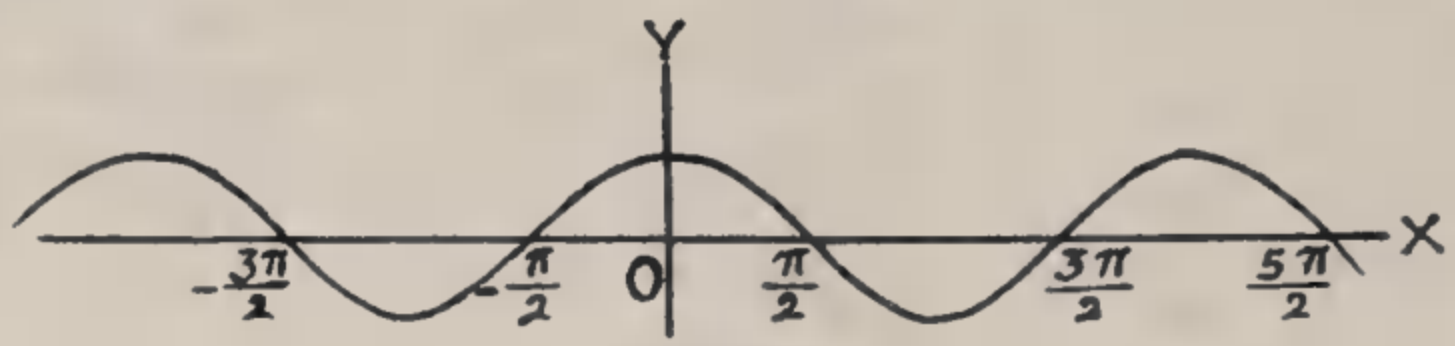
The curve represented by $y = \cos x$ is obtained in like manner and is quite similar to the sine curve. (See Fig. 17.)

To differentiate $\sin x$ we give to x the arbitrary values x_1, and $x_1 + \Delta x$ and compute for y the corresponding values $y_1 = \sin x_1$ and $y_1 + \Delta y = \sin (x_1 + \Delta x)$.



$y = \sin x$

FIG. 16



$y = \cos x$

FIG. 17

Subtracting y_1 from $y_1+\Delta y$, we have $\Delta y = \sin(x_1+\Delta x) - \sin x_1$. Dividing by Δx we have $\Delta y/\Delta x = \dfrac{\sin(x_1+\Delta x) - \sin x_1}{\Delta x}$.

To express the meaning of the above geometrically we may take a circle of unit radius and construct the angles x_1 and $(x_1+\Delta x)$, (Fig. 18). Then $M_1P_1 = \sin x_1$, $M_2P_2 = \sin(x_1+\Delta x)$, $QP_2 = \sin(x_1+\Delta x) - \sin x_1 = \Delta y$, and arc $P_1P_2 = \angle P_1P_2 = \Delta x$.

The limit approached by the ratio $\Delta y/\Delta x = QP_2/P_1P_2$ as $P_2 \to P_1$, or as $\Delta x \to 0$, is by previous definitions the sine of the angle $M_2P_2P_1$, since *in the limit* the arc P_1P_2 becomes a straight line, the hypotenuse of the right triangle P_1QP_2, which is similar to the right triangle P_1M_1O, whence angle $\angle M_2P_2P_1 = \angle x_1$.

In other words, $\lim\limits_{\Delta x \to 0} \dfrac{\Delta y}{\Delta x} = \lim\limits_{\Delta x \to 0} \dfrac{QP_2}{P_1P_2} = \dfrac{OM_1}{OP_1} = \cos x$, or $D_x \sin x = \cos x$.

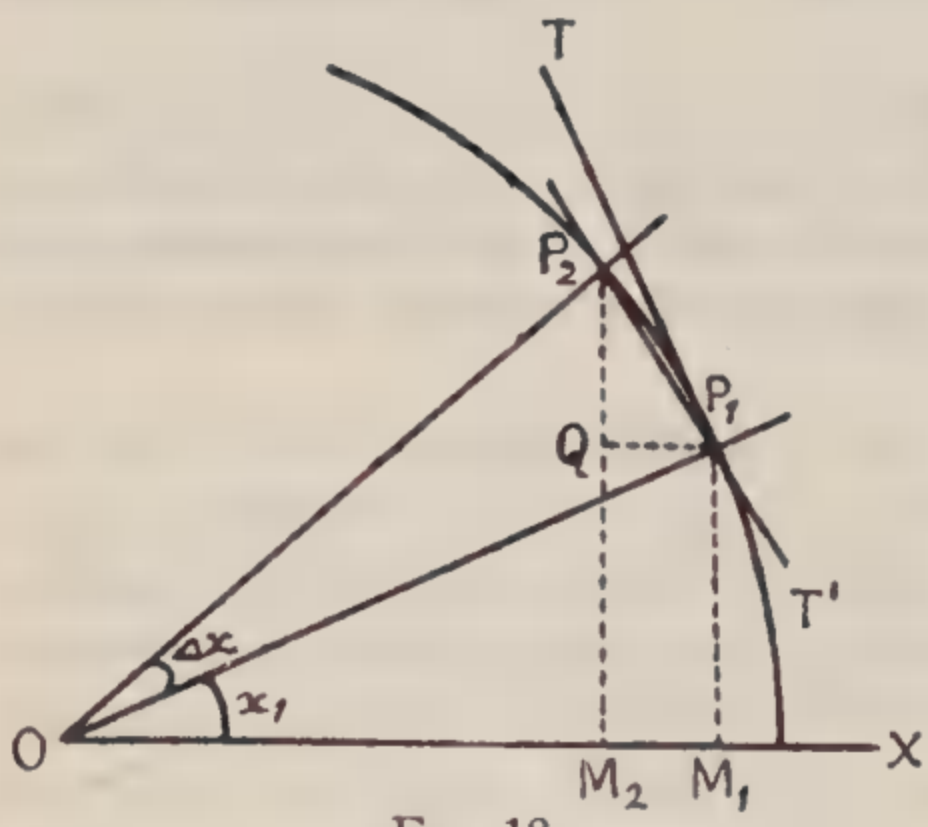


FIG. 18

It may be easily and similarly shown that $D_x \cos x = -\sin x$. Our main interest is in the second derivative. We see that the derivative of $\sin x$ is $\cos x$, and that the derivative of $\cos x$ is $-\sin x$.

Differentiating again we obtain the second derivative of $\sin x$ as $-\sin x$, and the second derivative of $\cos x$ as $-\cos x$.

The sine and cosine and their linear combinations are the only functions which when differentiated twice give us the second differential coefficient equal and of opposite sign to the original function. In symbols

$$\frac{d^2(\sin x)}{dx^2} = -\sin x; \text{ and } \frac{d^2(\cos x)}{dx^2} = -\cos x.$$

In physics we deal with many processes which are structurally periodic, which means that a definite physical condition constantly recurs after equal intervals of 'time'. The number of seconds or fractions of seconds within which the process runs its course is called the period. We know already that the simplest periodic functions are sine and cosine functions of the type

$$\sin(x+2n\pi)=\sin x, \text{ and}$$

$$\cos(x+2n\pi)=\cos x, \text{ where } n \text{ may have any integral value.}$$

Furthermore we have already seen that the first derivatives, therefore all derivatives of such functions are likewise simple sine and cosine functions. In particular, the second derivatives of the sine and cosine functions are likewise sine and cosine functions taken with the opposite algebraic sign.

If we express the variability of a process as a function of 'time', that is, by an equation of the form $S=F(t)$, then in a periodic process, $F(t_1+nT)=F(t_1)$, where T is the period and n any integer. If the process repeats itself, as in a periodic process, we must have

$$\left|\frac{dF}{dt}\right|_{t=t_1+nT}=\left|\frac{dF}{dt}\right|_{t=t_1} \text{ and } \left|\frac{d^2F}{dt^2}\right|_{t=t_1+nT}=\left|\frac{d^2F}{dt^2}\right|_{t=t_1},$$

but, as we have already seen, the sine and cosine functions satisfy these conditions.

A process which can be described by an equation of the type $S=A\sin\frac{2\pi t}{T}$ is called a *harmonic vibration* or, a 'pure sine vibration', or simply a 'vibration' or 'oscillation'. The constant A, which represents the maximum value of the displacement on either side, is called the amplitude. The period T is called the 'time of vibration', its reciprocal value which gives the number of vibrations in a unit of 'time' is called the vibration number or *frequency*.

As the second derivatives of sine and cosine functions are equal to the original functions taken with the opposite signs, we can describe harmonic vibrations by differential equations of the first degree (linear) and of the second order of the special type $\frac{d^2S}{dt^2}=-a^2S$, where $S=A\sin(\frac{2\pi t}{T}+\epsilon)$, A representing the amplitude, T the period, ϵ the phase of the vibration. The factor of proportionality a is taken as the square of any arbitrary real quantity to indicate that the right-hand side must always have the opposite sign to that of S.

The propagation of a vibration is called an advancing plane wave which has both velocity and direction.

Fourier has shown that any given form of wave may be represented by the superposition of a series of sine-waves, which gives sine-waves great theoretical and practical importance.

In writing this chapter I had two main aims. One was to briefly indicate the essential semantic factors involved in the differential methods. The other, to make the general reader and even specialists who are not mathematicians acquainted with some terms and rudiments of method which will be necessary for further discussion.

The differential methods involve semantic factors essential for a $\bar{A}$-system, the ∞-valued semantics of probability and for sanity and cannot be longer disregarded.

The main pressing issues are twofold. One, to formulate methods which would impart the $\bar{A}$ semantic reactions of the calculus, which need not involve any technicalities, and can be imparted in the most elementary home or school education. The other is to draw the attention of specialists to these semantic problems so that they will work them out.

An attempt to solve the first issue has been undertaken in the present volume. The second task will probably be accomplished in the not too distant future.

It is earnestly suggested to all scientists, professional men and teachers, who are not mathematicians, to become familiar with differential methods and so acquire the appropriate semantic reactions. Experience, in many cases, has shown that this will assist them in acquiring semantic balance and 'mental' efficiency. Teachers and physicians in particular, would be greatly helped in their efforts to train children and patients in the $\bar{A}$ reactions. The benefit is not in any 'calculations' whatsoever, but in the method and the related psycho-logical reactions.

There is an excellent, short, most elementary and amusing account of the calculus by Sylvanus P. Thompson *Calculus Made Easy* (Macmillan) which, for the present, is all that is needed for this purpose.